Exact and perturbative criteria for finite nonnegative linearization

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Abstract. For finite three-term recurrences normalized to have row sums one, we ask how far nonnegative linearization (property (P)), including products beyond the terminal degree, extends from the Krawtchouk arrays into the space of recurrence coefficients. Explicit coefficient bounds give relatively open neighborhoods with property (P) around every Krawtchouk array with bias $r < 1$ and around Szwarc's quadratic system, and these neighborhoods contain arrays outside every classical family. For fixed size $N \geq 2$ the largest admissible radius has exact order $(1-r)^N$ as r tends to the symmetric endpoint, where polynomial and suitably analytic deformation rules that keep property (P) must satisfy two first-order equalities. Criteria of another kind reach the endpoint: finite versions of Szwarc's inequalities control all products except those whose pairwise index sums all exceed the size, and a reversal identity settles these for palindromic systems. The reference positivity comes from a finite algebra with positive multiplication that occurs in Koelink's work.

Keywords. Finite orthogonal polynomials; nonnegative linearization; Krawtchouk polynomials; dual q-Krawtchouk polynomials; quadratic transformation; polynomial hypergroups.

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1. Introduction

Nonnegative linearization (property (P)) asks whether the product of two orthogonal polynomials is a nonnegative linear combination of polynomials in the same basis. For a finite system, the identity is understood on the orthogonality support. This distinction is essential when the formal product degree exceeds the terminal degree. In a normalization with $R_i(1) = 1$, nonnegative coefficients have sum one and define the product probabilities of a finite polynomial hypergroup.

The main subject of this paper is how far property (P) extends from the Krawtchouk arrays $K_{N,r}$ into the space of recurrence coefficients. Write $\rho_N(r)$ for the radius of the largest coefficient ball around $K_{N,r}$ in which every normalized array with $b_i \geq 0$ has property (P). For $0 < r < 1$, Theorem 3.2 gives the explicit bound $\rho_N(r) \geq \frac{1}{2}(\min(r, 1-r)/(32N))^N$, which is deliberately conservative: it is many orders of magnitude below the known upper

bounds (Remark 3.3), and the neighborhoods it admits contain arrays outside every classical family (Corollary 3.4); Theorem 7.6 gives explicit symmetric neighborhoods of Szwarc's quadratic system Q_n for $p_0 < 1/2$. Two regimes must be distinguished. For fixed $N \geq 2$ and $r \to 1$, $\rho_N(r)$ has exact order $(1-r)^N$ (Corollary 9.11), which the explicit bound attains up to a factor depending on N. For fixed $r < 1$ and $N \to \infty$, $\rho_N(r) \leq 3(1-r^2)^N$ [12, Corollary A.3], while the explicit bound decays faster than every exponential; the order in this regime is open.

At the symmetric endpoint $r = 1$ the corner coefficient $g(N, N; N)$ of $K_{N,1}$ vanishes (Lemma 9.1), and Section 9 shows that the obstruction to perturbative criteria there is structural: polynomial and suitably analytic deformation rules that keep property (P) at the symmetric Krawtchouk systems for all large N must satisfy two first-order equalities (Theorem 9.4 and Corollary 9.6), although each symmetric Krawtchouk array is a limit of arrays whose product coefficients are strictly positive on the whole triangle support (Proposition 9.8). Criteria that reach the endpoint are therefore of another kind. Section 8 adapts Szwarc's inequality method to finite systems. Under monotonicity conditions away from the terminal index, Theorem 8.1 controls every product except the deep triples, those in which each two indices have sum greater than N; under stronger conditions imposed through the terminal index, Theorem 8.3 controls all products. A reversal identity (Lemma 8.5) settles the deep triples for palindromic systems (Theorems 8.6 and 8.7), including the symmetric Krawtchouk array.

The explicit radius needs a reference with a positive margin. It comes from a finite algebra with positive multiplication (Sections 5 and 6), which gives property (P) whenever the profile $b_i + c_i$ has geometric increments and b_i/c_i is constant (Theorem 4.1). This family and its algebra are Koelink's [8, Theorems 3.1(ii) and 7.8]; for some parameters the arrays also come from Hamming and dual polar graphs [4, 5], see [12, Section 3], and [12, Theorem 3.1] contains all of them. We use the algebra for the margin of Lemma 3.1, and a graded version for Q_n in Section 7, and claim no priority for the algebra or the family.

Table 1 sorts the results into new criteria, previously known positivity and second proofs shared with [12], and says why each shared result appears here. From [12] this paper uses, in remarks only, the quadratic decomposition and the components of Q_n [12, Theorem 4.1 and Corollary 4.2], and it cites for comparison the bound [12, Corollary A.3] and the failures of convex combinations [12, Section 5]. No written proof depends on [12]; two Lean proofs do, as stated in the section on code and formalization.

The positive Krawtchouk range is classical [7, 6, 20]; Corollary B.1 derives our endpoint normalization directly.[1] Szwarc's finite-system problem [20, §5, Problem 2] motivates the present criteria. Earlier recurrence criteria include Askey's finite result for products of degree sum at most N [1, Part II, Theorem 1 and Corollary 1], Szwarc's discrete hyperbolic characterization [19, Theorem 1], his monotonicity criteria for infinite systems [20, Theorems 3.2–3.4], from which Section 8 starts, and the chain-sequence criteria of Młotkowski and Szwarc [11] for infinite systems whose orthogonality measure is supported on a conver-

[1] The linked author version of [20], §5, Problem 2, printed p. 125, uses inconsistent parameter conventions: its displayed weights have mean $N(1-p)$, whereas its degree-zero recurrence gives $K_1(x) = (x - pN)/((1-p)N)$, so that the displayed weights give $\int K_1 \, d\mu = (1-2p)/(1-p)$.

Table 1: The results of this paper by status, with the reason why the results shared with [12] appear here.

Results	Status	Role here
Thm. 3.2, Cor. 3.4, Lem. 7.5	new	explicit neighborhoods of $K_{N,r}$
Thm. 7.6, Cor. 7.7	new	neighborhoods of Q_n
Thms. 8.1, 8.3, Cors. 8.2, 8.4	new	inequality criteria
Lem. 8.5, Thms. 8.6, 8.7, Cor. 8.8	new; case $t = N$ of the lemma in [3]	deep triples, symmetric endpoint
Section 9 except Lems. 9.1, 9.5	new	endpoint obstruction, order of $\rho_N(r)$
Thm. 4.1, Cors. B.1, B.2	known [7, 6, 20, 8]; also from [12, Thm. 3.1]	reference positivity; its algebra gives the margin of Lem. 3.1
Thm. 7.1	stated by Szwarc [20], proved in [12, Thm. 4.3]; second proof	its graded algebra gives the margin of Lem. 7.4
Lem. 9.1	also in [12]	corner coefficient at $r = 1$
Lem. 9.5	classical Nörlund–Rice estimate	analytic rules

gent sequence of points. Szwarc's quadratic-transformation arguments [17, 18] pass from symmetric infinite systems to their even parts; the converse direction for finite systems is [12, Theorem 4.1]. Miclo [9, Proposition 3.3] characterizes the hypergroup property of a finite birth–death chain at a boundary point spectrally, and Terwilliger [21, Problems 36.4 and 36.5] and Blau and Hein [2, Theorem 1.1] study related classification questions for finite systems.

The companion paper [12] proves the log-concave Newton criterion, and [13] treats varying ratios. Here Section 2 fixes conventions and Section 3 proves the explicit neighborhoods; Sections 4–6 give the reference positivity and the margin; Section 7 treats Q_n, Section 8 the inequality and reversal criteria, and Section 9 the endpoint. Section 10 states open problems, Appendix A collects the exact counterexamples to shape conditions, separately from numerical evidence, and Appendix B derives the classical positivity results and compares the geometric family with Koelink's construction.

2. Conventions

Consider real coefficients indexed by $0 \le i \le N$, where $N \ge 1$, satisfying

$$\begin{gathered} a_0 = 1, \quad b_0 = c_0 = 0, \quad a_N = 0, \quad a_i + b_i + c_i = 1, \\ a_i > 0 \ (i < N), \qquad c_i > 0 \ (i > 0). \end{gathered} \tag{2.1}$$

Set $R_{-1} = 0$, $R_0 = 1$, and determine $R_1, \dots, R_N$ by

$$xR_i = a_i R_{i+1} + b_i R_i + c_i R_{i-1} \qquad (0 \le i < N). \tag{2.2}$$

At $i = 0$, (2.2) gives $R_1 = x$; induction at $x = 1$, using the row sums and $a_i > 0$ for $i < N$, gives $R_i(1) = 1$ for $0 \le i \le N$. The terminal relation is $(x - b_N)R_N - c_N R_{N-1} = 0$ on the orthogonality support. Linearization means equality of functions on this support, rather than an unrestricted polynomial identity when the degree sum exceeds N. We call an array satisfying (2.1) *normalized*. Its recurrence matrix has row sums one, but (2.1) allows $b_i < 0$; we call the array *stochastic* only when all $b_i \ge 0$, so that the matrix is stochastic. For $0 < r \le 1$ the *Krawtchouk array* $K_{N,r}$ has $a_i = (N-i)/N$, $b_i = (1-r)i/N$ and $c_i = ri/N$; it has property (P), see Corollary B.1.

Write the tridiagonal recurrence matrix with the row convention

$$J_{i,i+1} = a_i, \qquad J_{i,i} = b_i, \qquad J_{i,i-1} = c_i,$$

with all other entries zero and indices outside $0, \dots, N$ omitted.

Lemma 2.1 *(Finite recurrence realization and triangle support).* *Every recurrence satisfying (2.1) has a positive orthogonality measure on exactly $N+1$ real points. If J is its tridiagonal recurrence matrix, then its finite product coefficients satisfy*

$$g(m, n; k) = [R_m(J)]_{n,k}. \tag{2.3}$$

They vanish unless $|m - n| \le k \le \min(m + n, N)$, regardless of their signs inside this interval.

Proof. Choose positive π_i with $\pi_0 = 1$ and $\pi_{i+1} = \pi_i a_i / c_{i+1}$. For $D = \operatorname{diag}(\sqrt{\pi_i})$, $H = DJD^{-1}$ is real symmetric. Induction in (2.2) gives the row-vector identity $e_0^{\mathsf{T}} R_i(J) = e_i^{\mathsf{T}}$, hence $e_0^{\mathsf{T}} R_i(H) = \pi_i^{-1/2} e_i^{\mathsf{T}}$. In particular e_0 is cyclic for H. In an orthogonal eigenspace decomposition of H, the cyclic span of e_0 has dimension at most the number of distinct eigenvalues; here it is all of $\mathbb{R}^{N+1}$, so H has $N+1$ distinct eigenvalues and the projection of e_0 on each eigenspace is nonzero. The spectral measure μ of H at e_0 therefore has positive mass at each of $N+1$ real points, and

$$\int R_i R_j \, d\mu = e_0^{\mathsf{T}} R_i(H) R_j(H) e_0 = \mathbf{1}_{i=j} \pi_i^{-1}.$$

The terminal relation holds on this support: its polynomial $Q(x) = (x - b_N)R_N - c_N R_{N-1}$ satisfies $e_0^{\mathsf{T}} Q(J) = 0$. Since e_0 is cyclic and $Q(J)$ commutes with J, $Q(J) = 0$. Multiplying a basis polynomial by x acts by the corresponding row of J, including the terminal row. Induction on powers of x therefore gives (2.3). Orthogonality also gives

$$g(m, n; k) = \frac{\int R_m R_n R_k \, d\mu}{\int R_k^2 \, d\mu}.$$

If one of m, n, k exceeds the sum of the other two, its polynomial is orthogonal to their product, whose degree is smaller. This proves the asserted support. ∎

3. Explicit perturbation neighborhoods

Lemma 3.1 *(A uniform positive margin in the asymmetric case).* *For the Krawtchouk array $K_{N,r}$ with $0 < r < 1$, put $s = \min(r, 1-r)$. Every coefficient in its triangle support obeys*

$$g(m,n;k) \geq (s/4)^N \quad \text{if } |m-n| \leq k \leq \min(m+n, N). \tag{3.1}$$

This lemma comes from a single positive term in the finite algebra of Section 5; its proof is given at the end of Section 6.

Theorem 3.2 *(Explicit perturbation inequalities).* *Let $N \geq 1$, $0 < r < 1$, $s = \min(r, 1-r)$, and*

$$\eta_N(r) = \frac{1}{2}\left(\frac{s}{32N}\right)^N. \tag{3.2}$$

Suppose coefficients $\widetilde{a}_i, \widetilde{b}_i, \widetilde{c}_i$ satisfy (2.1), with $\widetilde{b}_i \geq 0$ for every i, and

$$\max_{0\leq i\leq N}\left\{\left|\widetilde{a}_i - \frac{N-i}{N}\right|, \left|\widetilde{b}_i - \frac{(1-r)i}{N}\right|, \left|\widetilde{c}_i - \frac{ri}{N}\right|\right\} \leq \eta_N(r). \tag{3.3}$$

Then the finite system has nonnegative linearization. Throughout its triangle support,

$$\widetilde{g}(m,n;k) \geq \tfrac{1}{2}(s/4)^N.$$

The product coefficients sum to one.

Proof. Use tildes for the perturbed system and untilded letters for its $q = 1$ reference at r. Let $\eta = \eta_N(r)$ and use the maximum absolute row-sum matrix norm. Both J and $\widetilde{J}$ are stochastic matrices, since $b_i \geq 0$ and $\widetilde{b}_i \geq 0$, so $\|J\|_\infty = \|\widetilde{J}\|_\infty = 1$ and $\|\widetilde{J} - J\|_\infty \leq 3\eta$. Put $B_m = R_m(J)$, $\widetilde{B}_m = \widetilde{R}_m(\widetilde{J})$ and $e_m = \|\widetilde{B}_m - B_m\|_\infty$. Theorem 4.1 at $q = 1$ and (2.3) give $\|B_m\|_\infty = 1$ for all $m \leq N$. Also $e_0 = 0$ and $e_1 \leq 3\eta$.

Subtract the two matrix recurrences for $1 \leq m < N$. Writing $\Delta_m = \widetilde{B}_m - B_m$ gives exactly

$$\begin{aligned}\widetilde{a}_m\Delta_{m+1} = {} & (\widetilde{J} - \widetilde{b}_m I)\Delta_m - \widetilde{c}_m\Delta_{m-1} \\ & + \big(\widetilde{J} - J - (\widetilde{b}_m - b_m)I\big)B_m \\ & - (\widetilde{c}_m - c_m)B_{m-1} - (\widetilde{a}_m - a_m)B_{m+1}.\end{aligned}$$

Since $\eta \leq 1/(2N)$, $\widetilde{a}_m \geq 1/(2N)$. The terms not containing a Δ have combined norm at most 6η. Also $0 \leq \widetilde{b}_m, \widetilde{c}_m \leq 1$, so

$$e_{m+1} \leq 4Ne_m + 2Ne_{m-1} + 12N\eta. \tag{3.4}$$

Set $L = 8N$. Induction gives $e_m \leq \eta L^m$ for $1 \leq m \leq N$: the base case follows from $3 \leq L$, and, for $m \geq 1$,

$$4N + \frac{2N}{L} + \frac{12N}{L^m} \leq 4N + \frac{1}{4} + \frac{3}{2} \leq 8N = L.$$

Thus every matrix entry differs from its reference by at most

$$\eta(8N)^N = \frac{1}{2}(s/4)^N.$$

Lemma 3.1 gives the positive lower bound on triangle entries, while Lemma 2.1 makes all entries outside the triangle exactly zero. Finally $\widetilde{R}_m(1) = 1$ follows by recurrence from the row sums $\widetilde{a}_i + \widetilde{b}_i + \widetilde{c}_i = 1$, so evaluating each product at 1 gives coefficient sum one. Here 1 belongs to the support because $\widetilde{J}\mathbf{1} = \mathbf{1}$ and Lemma 2.1 identifies that support with its spectrum. No product-positivity inequality for the perturbed system has been assumed. ■

Remark 3.3 (Size of the radius). The radius $\eta_N(r)$ shows that a neighborhood exists and how it scales as $r \to 1$; it is not meant to approximate the stability radius $\rho_N(r)$, the radius of the largest admissible coefficient ball around $K_{N,r}$ with property (P). At $r = \frac{1}{2}$, compared with the upper bound $\rho_N(\frac{1}{2}) \le 3(\frac{3}{4})^N$, valid for $N \ge 16$ [12, Corollary A.3]:

N	5	10	20	40	120
$\eta_N(\frac{1}{2})$	$1.5 \cdot 10^{-13}$	$4.3 \cdot 10^{-29}$	$3.6 \cdot 10^{-63}$	$2.3 \cdot 10^{-137}$	$2.9 \cdot 10^{-467}$
$3(\frac{3}{4})^N$	—	—	$9.5 \cdot 10^{-3}$	$3.0 \cdot 10^{-5}$	$3.1 \cdot 10^{-15}$

The gap comes from the two crude steps of the proof: the single-term margin $(s/4)^N$ of Lemma 3.1 and the worst-case amplification $(8N)^N$ of the error recurrence.

Corollary 3.4 *(Finite systems outside the geometric family).* *For every $N \ge 2$, take the Krawtchouk coefficients at $r = 1/2$, and replace only*

$$a_1 \longmapsto a_1 - h, \qquad b_1 \longmapsto b_1 + h, \qquad 0 < h \le \frac{1}{4}(64N)^{-N}.$$

These recurrences have nonnegative linearization and do not satisfy the constant-ratio hypothesis of Theorem 4.1. More generally, Theorem 3.2 admits a relatively open neighborhood of the asymmetric Krawtchouk array in the affine space defined by the row-sum and boundary equalities in (2.1).

Proof. All initial and terminal conventions and the row sums are preserved. The bound on h is half the radius (3.2) at $r = 1/2$, and is smaller than $a_1 = (N-1)/N$, so all required signs hold. The perturbation theorem applies. At index 1, $b_1/c_1 = 1 + 2Nh > 1$, whereas at index N the ratio remains 1; these indices differ because $N \ge 2$. Thus this array lies outside the geometric family. At the asymmetric reference, all free coefficients constrained to be positive are strictly positive; the strict version of (3.3) therefore contains a relatively open neighborhood in that affine space; the sign conditions define the admissible region within it. ■

4. The geometric reference family

The margin of Lemma 3.1 comes from a finite algebra with positive multiplication. It proves property (P) for a family of arrays that contains $K_{N,r}$; the family and the algebra occur in

Koelink's work [8] (Proposition B.3), and [12, Theorem 3.1] contains the family. For integers $j \geq 0$ put $[j]_q = 1 + q + \cdots + q^{j-1}$, with $[0]_q = 0$; in particular $[j]_1 = j$.

Theorem 4.1 *(Geometric increment criterion).* *Assume (2.1). Set $d_i = b_i + c_i$. Suppose*

1. *$b_i/c_i = \beta$ for every $1 \leq i \leq N$, where $\beta \geq 0$;*
2. *$\delta_i = d_i - d_{i-1} > 0$ for $1 \leq i \leq N$;*
3. *for some $q \geq 1$, $\delta_{i+1} = q\delta_i$ whenever $1 \leq i < N$.*

Then the recurrence has a positive orthogonality measure on exactly $N+1$ distinct real points. Its normalized polynomials satisfy

$$R_m R_n = \sum_{k=0}^{N} g(m,n;k) R_k, \qquad g(m,n;k) \geq 0, \qquad \sum_{k=0}^{N} g(m,n;k) = 1 \tag{4.1}$$

on that support for all $0 \leq m, n \leq N$.

Lemma 4.2 *(Coefficient parametrization).* *The hypotheses of Theorem 4.1 are equivalent to*

$$a_i = \frac{q^i [N-i]_q}{[N]_q}, \qquad b_i = (1-r)\frac{[i]_q}{[N]_q}, \qquad c_i = r\frac{[i]_q}{[N]_q} \tag{4.2}$$

for $q \geq 1$, $0 < r \leq 1$. When $N = 1$, one may choose $q = 1$.

Proof. The terminal sum gives $d_N = 1$, and $d_0 = 0$. Geometric increments give $d_i = \delta_1 [i]_q$, hence $\delta_1 = 1/[N]_q$. The constant ratio gives $c_i = d_i/(1+\beta)$ and $b_i = \beta d_i/(1+\beta)$. Put $r = (1+\beta)^{-1}$. Finally $[N]_q - [i]_q = q^i [N-i]_q$. These calculations reverse to give the converse, including $q = 1$ and $N = 1$. ∎

The parameter conventions used below are

q	$q \geq 1$ (geometric increment ratio)
β	$b_i/c_i \geq 0$
r	$1/(1+\beta) \in (0,1]$
p	$1/r = 1 + \beta \geq 1$
p_0	$r/(1+r) \in (0, 1/2]$
symmetric Krawtchouk case	$r = 1, \quad p_0 = 1/2$

5. A finite algebra with positive multiplication

Fix $p \geq 1$ and $q \geq 1$. Starting with $A_{N+1} = \mathbb{R}$, form recursively the commutative real algebras

$$A_j = A_{j+1}[z_j] \Big/ \left(z_j^2 - \left((q-1) \sum_{l=j+1}^{N} z_l + p - 1 \right) z_j - p q^{N-j} \right). \tag{5.1}$$

Write $A = A_1$. For $S \subseteq \{1, \ldots, N\}$ let $z_S = \prod_{j \in S} z_j$, with $z_\varnothing = 1$.

Lemma 5.1 *(Basis and positivity).* *The elements z_S form a basis of A, and all their multiplication coefficients are nonnegative.*

Proof. Division by a monic quadratic over any commutative coefficient ring gives a unique remainder of degree at most one. Applying this successively in (5.1) gives the asserted basis and dimension 2^N. For positivity, replace a square by

$$z_j^2 \;=\; (q-1)\sum_{l>j} z_j z_l + (p-1)z_j + pq^{N-j}.$$

Every displayed coefficient is nonnegative. This rewriting terminates: give z_j weight 2^{N-j} and give a monomial the sum of the weights of its factors, counted with multiplicity. Each replacement strictly decreases this nonnegative integer weight, because $l > j$ implies $2^{N-l} < 2^{N-j}$. The final expression is in the unique squarefree basis, proving the claim. ∎

Lemma 5.2 *(Positive associative inner product).* *Let τ extract the coefficient of $z_\varnothing$. Then*

$$\tau(z_S z_T) \;=\; \mathbf{1}_{S=T}\, p^{|S|} q^{\sum_{j\in S}(N-j)}. \tag{5.2}$$

Consequently $\langle u, v\rangle = \tau(uv)$ is positive definite, and multiplication by every element of A is self-adjoint.

Proof. Induct downwards through the tower. If j belongs to exactly one of S, T, their product has the form $z_j v$ with $v \in A_{j+1}$ and has zero constant coefficient in z_j. If it belongs to both, the constant coefficient in z_j is $pq^{N-j} z_{S\setminus\{j\}} z_{T\setminus\{j\}}$ by (5.1). If it belongs to neither, pass to A_{j+1}. Iteration proves (5.2). All diagonal factors are positive. Associativity and commutativity give $\tau((hu)v) = \tau(u(hv))$. ∎

Put $E_k = \sum_{|S|=k} z_S$ for $0 \le k \le N$, and set $E_{-1} = E_{N+1} = 0$.

Lemma 5.3 *(Three-term multiplication).* *For $0 \le k \le N$,*

$$E_1 E_k \;=\; [k+1]_q E_{k+1} + (p-1)[k]_q E_k + pq^{k-1}[N-k+1]_q E_{k-1}. \tag{5.3}$$

The last term is interpreted as zero for $k = 0$.

Proof. For $k = 0$ the identity reads $E_1 = E_1$. For $N = 1, k = 1$ it is exactly $z_1^2 = (p-1)z_1 + p$. For the induction on N, write F_j for the elementary squarefree sums in $z_2, \dots, z_N$, with $F_j = 0$ outside $0 \le j \le N-1$. Then $E_k = F_k + z_1 F_{k-1}$ and $E_1 = F_1 + z_1$. For $1 \le k \le N-1$, use (5.1) to obtain

$$E_1 E_k \;=\; F_1 F_k + z_1 F_k + q z_1 F_1 F_{k-1} + (p-1) z_1 F_{k-1} + pq^{N-1} F_{k-1}.$$

Apply the induction hypothesis to $F_1 F_k$ and $F_1 F_{k-1}$. The coefficients of F_{k+1} and $z_1 F_k$ are respectively $[k+1]_q$ and $1 + q[k]_q = [k+1]_q$. The coefficients of F_k and $z_1 F_{k-1}$ are both $(p-1)[k]_q$, since $1 + q[k-1]_q = [k]_q$. The coefficient of F_{k-1} is

$$pq^{k-1}[N-k]_q + pq^{N-1} \;=\; pq^{k-1}[N-k+1]_q,$$

which is also the coefficient of z_1F_{k-2}. Regrouping gives (5.3). For the endpoint $k = N$, $E_N = z_1F_{N-1}$, so the same square relation gives

$$E_1E_N = qz_1F_1F_{N-1} + (p-1)z_1F_{N-1} + pq^{N-1}F_{N-1}.$$

The tail endpoint identity is

$$F_1F_{N-1} = (p-1)[N-1]_qF_{N-1} + pq^{N-2}F_{N-2}.$$

Substitution gives $(p-1)[N]_qE_N + pq^{N-1}E_{N-1}$, as required. This separately checks the terminating boundary. ■

Proposition 5.4 *(Positive subalgebra).* *The space $B = \operatorname{span}\{E_0, \dots, E_N\}$ is the subalgebra generated by E_1. Every product E_mE_n has nonnegative coefficients in this basis.*

Proof. All E_k are independent, since their squarefree supports are nonempty and disjoint. Equation (5.3) shows that B is invariant under multiplication by E_1. Since $[k+1]_q > 0$, its interior recurrence also expresses each E_{k+1} as a polynomial in E_1. Thus $B = \mathbb{R}[E_1]$, proving closure under multiplication, including products of total index greater than N. Expand such a product as $\sum_k h_kE_k$. Its coefficient at each squarefree monomial z_S is $h_{|S|}$. Lemma 5.1 makes that coefficient nonnegative. There is a subset of each cardinality $0, \dots, N$, so every h_k is nonnegative. ■

6. Normalization and the finite orthogonal system

Lemma 6.1 *(Positive normalization).* *Evaluation $z_j \mapsto pq^{N-j}$ defines an algebra character ϵ of A. Put $t_k = \epsilon(E_k) > 0$. Then*

$$t_0 = 1, \quad t_1 = p[N]_q, \qquad \frac{t_{k+1}}{t_k} = pq^k\frac{[N-k]_q}{[k+1]_q} \quad (0 \le k < N). \tag{6.1}$$

Proof. For $h = N - j$, substitution in the right-hand side of (5.1) gives

$$(q-1)p^2q^h[h]_q + (p-1)pq^h + pq^h = p^2q^h\big((q-1)[h]_q + 1\big) = p^2q^{2h}.$$

Thus all defining relations hold. The generating polynomial of the t_k is $T(u) = \prod_{j=0}^{N-1}(1 + pq^ju)$, whose coefficients are strictly positive. When $q > 1$, the identity $(1+pu)T(qu) = (1+pq^Nu)T(u)$ gives $(q^k-1)t_k = p(q^N - q^{k-1})t_{k-1}$ for $1 \le k \le N$, which is (6.1). When $q = 1$, $t_k = \binom{N}{k}p^k$ and the same ratio follows directly. ■

Proof of Theorem 4.1. Use Lemma 4.2 and put $p = 1/r \ge 1$. Construct A, B as above and set $X = E_1/t_1$, $U_k = E_k/t_k$. Divide (5.3) by t_1t_k and use (6.1). The three coefficients are

$$\begin{aligned}
\frac{[k+1]_qt_{k+1}}{p[N]_qt_k} &= \frac{q^k[N-k]_q}{[N]_q},\\
\frac{(p-1)[k]_q}{p[N]_q} &= (1-r)\frac{[k]_q}{[N]_q},\\
\frac{pq^{k-1}[N-k+1]_qt_{k-1}}{p[N]_qt_k} &= r\frac{[k]_q}{[N]_q}.
\end{aligned}$$

At $k = 0$ omit the third expression; at $k = N$ omit the first. Consequently $U_k = R_k(X)$ for every $k \le N$, with the stated terminal relation.

The orthogonality support follows from the real spectral theorem in finite dimensions. On B use the positive inner product from Lemma 5.2. Multiplication by X is self-adjoint. The vector $1 = E_0$ is cyclic because the E_k are polynomials of successive degrees in X and form a basis. In an orthogonal eigenspace decomposition of a self-adjoint operator, the cyclic span of one vector has dimension at most the number of distinct eigenvalues. Here the cyclic span has dimension $N + 1$, so there are exactly $N + 1$ distinct eigenvalues, and the projection of 1 on every eigenspace is nonzero. Give each eigenvalue the squared norm of that projection as its mass. All masses are positive, sum to $\langle 1, 1\rangle = 1$, and yield

$$\int f(x)h(x)\,d\mu(x) \;=\; \langle f(X), h(X)\rangle$$

for real polynomials f, h, identifying an algebra element with its action on 1. The E_k are orthogonal by (5.2), so the R_k are orthogonal for this measure.

Proposition 5.4 and positive t_k give nonnegative product coefficients for the U_k. Applying the character ϵ gives their sum equal to one because $\epsilon(U_k) = 1$. Polynomial identities in the algebra hold on every spectral point; conversely the R_k, having successive degrees at most N, form a basis of functions on the $N + 1$ distinct points. This proves uniqueness and (4.1) for all products. ∎

Proof of Lemma 3.1. Specialize the algebra of Section 5 to $q = 1$, and put $y_i = z_i/p$, where $p = 1/r$. Then $y_i^2 = (1 - r)y_i + r$, and $U_m = \binom{N}{m}^{-1} \sum_{|S|=m} y_S$. The reduction of a product $y_S y_T$ has nonnegative coefficients, each chosen term being a product of at most N factors from $\{r, 1 - r\}$. Its coefficient is therefore at least s^N. For each admissible k, at least one such term has degree k. Assume $m \ge n$. If $k \le m$, take $T \subseteq S$, with sizes n, m, and retain the linear term for exactly $k - (m - n)$ overlap coordinates. This number lies between 0 and n. If $k \ge m$, choose sets with overlap $m + n - k$ and union of size $k \le N$, and retain the linear term at every overlap coordinate. Permutation symmetry makes the coefficient of y_U depend only on $|U| = k$. Hence the coefficient in the normalized elementary-symmetric basis is at least

$$\frac{s^N \binom{N}{k}}{\binom{N}{m}\binom{N}{n}} \;\ge\; \frac{s^N}{4^N}.$$

This uses a single positive term and a denominator bound, not an explicit formula for the product coefficients. ∎

7. Szwarc's quadratic system

Problem 2 of [20, §5] does not stop at the Krawtchouk recurrence. With his parameter written as p_0, Szwarc introduces the symmetric recurrence of length $2N + 1$

$$yQ_{2n} \;=\; (N - n)Q_{2n+1} + nQ_{2n-1}, \qquad yQ_{2n-1} \;=\; (1 - p_0)Q_{2n} + p_0 Q_{2n-2}, \tag{7.1}$$

records that $Q_{2n}(y)$ is the Krawtchouk polynomial $K_n(y^2)$ of his recurrence, states that the Q_n have nonnegative linearization for $p_0 \le 1/2$, and asks for a general criterion that would

handle this case. The companion paper [12, Section 4] explains why his Theorem 3.3 for infinite symmetric systems does not apply, and proves this positivity through a quadratic decomposition of symmetric finite systems. This section gives a second proof, by a graded version of the algebra of Section 5, which also yields an explicit coefficient criterion for symmetric finite systems near (7.1) (Theorem 7.6).

Put $M = 2N$, $R_{2n}(t) = Q_{2n}(\sqrt{N}t)$ and $R_{2n-1}(t) = \sqrt{N}\,Q_{2n-1}(\sqrt{N}t)$. Substitution in (7.1) shows that the R_k satisfy (2.2), with N replaced by M, for the array

$$b_k = 0, \qquad (a_{2n}, c_{2n}) = \Big(\frac{N-n}{N}, \frac{n}{N}\Big), \qquad (a_{2n-1}, c_{2n-1}) = (1-p_0,\, p_0). \tag{7.2}$$

It satisfies (2.1) with M in place of N whenever $0 < p_0 < 1$. The two normalizations differ by positive factors and a positive dilation, so they have the same product coefficients up to positive factors.

Theorem 7.1 *(Szwarc's quadratic system).* *Let $N \geq 1$ and $0 < p_0 \leq 1/2$. The array (7.2) has a symmetric positive orthogonality measure on $2N+1$ points, and for all $0 \leq m, n \leq 2N$*

$$R_m R_n = \sum_{k=0}^{2N} g(m,n;k) R_k, \qquad g(m,n;k) \geq 0, \qquad \sum_k g(m,n;k) = 1$$

on that support. If $1/2 < p_0 < 1$, then $g(2,2;2) < 0$.

Throughout this section put $r = p_0/(1-p_0)$, so that $0 < r \leq 1$ exactly when $p_0 \leq 1/2$. Let A be the commutative algebra generated by $y_1, \dots, y_N$ subject to $y_j^2 = (1-r)y_j + r$. This is the algebra of Section 5 with $q = 1$ after the rescaling $y_j = z_j/p$ used in Lemma 3.1. Its squarefree monomials y_S form a basis, and for $r \leq 1$ all multiplication coefficients are nonnegative. Call an element of A nonnegative if its coefficients in this basis are nonnegative.

Lemma 7.2 *(Edge idempotents).* *For $1 \leq l \leq N$ put $f_l = (r + y_l)/(1+r)$. Then $y_l f_l = f_l$ and $f_l^2 = f_l$. The ideal $f_l A$ has basis $f_l y_S$, $S \subseteq \{1, \dots, N\} \setminus \{l\}$, and $f_l u$ has nonnegative coefficients in this basis whenever $u \in A$ is nonnegative.*

Proof. First $(1+r) y_l f_l = r y_l + (1-r) y_l + r = (1+r) f_l$. Then $(1+r) f_l^2 = r f_l + y_l f_l = (1+r) f_l$. Let A' be generated by the y_j with $j \neq l$, so that $A = A' \oplus y_l A'$. Since $f_l y_l = f_l$, $f_l A = f_l A'$. The map $u \mapsto f_l u = (ru + y_l u)/(1+r)$ is injective on A' by the direct sum, which gives the basis. For the last claim, $f_l y_S = f_l y_{S \setminus \{l\}}$ for every S. ∎

Let $V = A \oplus \bigoplus_{l=1}^N f_l A$, on which A acts by multiplication in every summand, and call a vector nonnegative if all of its coordinates are nonnegative in the bases above. Define the A-linear operator

$$Y(u; v_1, \dots, v_N) = \Big(\frac{1}{N}\sum_{l=1}^N v_l\,;\, f_1 u, \dots, f_N u\Big).$$

Put $U_{2n} = \binom{N}{n}^{-1} \sum_{|S|=n} y_S \in A$ for $0 \leq n \leq N$. For $0 \leq n \leq N-1$, let U_{2n+1} be the vector in $\bigoplus_l f_l A$ whose lth coordinate is $\binom{N-1}{n}^{-1} \sum f_l y_S$, summed over $S \subseteq \{1, \dots, N\} \setminus \{l\}$ with $|S| = n$.

Lemma 7.3 *(Three-term action).* *With $U_{-1} = U_{2N+1} = 0$ and the coefficients (7.2), $YU_k = a_k U_{k+1} + c_k U_{k-1}$ for $0 \le k \le 2N$.*

Proof. The lth coordinate of $Y\sum_{|S|=n} y_S$ is $\sum_{S\not\ni l} f_l y_S + \sum_{S'\not\ni l} f_l y_{S'}$, with $|S| = n$ and $|S'| = n-1$, because $f_l y_S = f_l y_{S\setminus\{l\}}$. Dividing by $\binom{N}{n}$ and using $\binom{N-1}{n}/\binom{N}{n} = (N-n)/N$ and $\binom{N-1}{n-1}/\binom{N}{n} = n/N$ gives the even case. For the odd case, $(1+r)f_l y_S = r y_S + y_{S\cup\{l\}}$ when $l \notin S$. Each n-set occurs for $N-n$ choices of l and each $(n+1)$-set for $n+1$ choices, so

$$YU_{2n+1} = \frac{r(N-n)\binom{N}{n}U_{2n} + (n+1)\binom{N}{n+1}U_{2n+2}}{N(1+r)\binom{N-1}{n}} = \frac{rU_{2n} + U_{2n+2}}{1+r}.$$

Finally $r/(1+r) = p_0$ and $1/(1+r) = 1-p_0$. ∎

Proof of Theorem 7.1. Lemma 2.1 gives a positive measure on the $2N+1$ zeros of the terminal polynomial $T(t) = tR_{2N}(t) - R_{2N-1}(t)$. Since $b \equiv 0$, the matrix J is conjugate to $-J$ by $\mathrm{diag}((-1)^i)$, which fixes e_0; hence the measure is symmetric and $R_k(-t) = (-1)^k R_k(t)$. Put $\Phi(f) = f(Y)U_0$, where $U_0 = (1;0,\dots,0)$. Lemma 7.3 gives $\Phi(R_k) = U_k$ for $k \le 2N$ and $\Phi(T) = YU_{2N} - U_{2N-1} = 0$. Every polynomial vanishing on the support is a multiple of T, so Φ maps it to zero. Hence the product identity on the support becomes

$$R_m(Y)R_n(Y)U_0 = \sum_k g(m,n;k)U_k.$$

The supports of the U_k in the bases of V are nonempty and disjoint: U_{2n} uses the y_S with $|S| = n$, and U_{2n+1} uses the $f_l y_S$ with $l \notin S$, $|S| = n$. Each basis vector lies in exactly one of them, with positive coefficient. Consequently $g(m,n;k) \ge 0$ for every k whenever the left side is nonnegative.

Suppose m is even. Since polynomials in Y commute and Y is A-linear,

$$R_m(Y)R_n(Y)U_0 = R_n(Y)(U_m U_0) = U_m\, R_n(Y)U_0 = U_m U_n.$$

Here $U_m \in A$ and U_n are nonnegative, so their product is nonnegative by the positivity of A and Lemma 7.2. The same holds when n is even, since $g(m,n;k) = g(n,m;k)$. If m,n are odd and k is odd, then $g(m,n;k) = 0$ by symmetry of the measure. If m,n are odd and k is even, then $g(m,n;k)\int R_k^2\,d\mu = \int R_m R_n R_k\,d\mu = g(k,m;n)\int R_n^2\,d\mu \ge 0$. The point $t = 1$ belongs to the support because $J\mathbf{1} = \mathbf{1}$, and $R_k(1) = 1$, so the coefficients sum to one.

For $1/2 < p_0 < 1$ we have $r > 1$. The identities above are purely algebraic and remain valid, but A is no longer positive. In particular $U_2^2 = N^{-2}\big((1-r)\sum_j y_j + Nr + 2\sum_{i<j} y_i y_j\big)$, whose U_2 coefficient $g(2,2;2) = (1-r)/N$ is negative. ∎

Since Y^2 acts on A by multiplication by $N^{-1}\sum_l f_l = (r+X)/(1+r)$, with $X = N^{-1}\sum_l y_l$, the even part of the theorem is Corollary B.1: $R_{2n}(t) = K_n\big((1+r)t^2 - r\big)$ in the normalization of that corollary. Thus Theorem 7.1 recovers the full positive Krawtchouk range by Szwarc's quadratic route. At $p_0 = 1/2$ the operator Y is the normalized adjacency operator of the subdivided N-cube seen from a vertex, and the theorem reduces to counting in that graph.

Lemma 7.4 *(Margin for the quadratic system).* *Let $N \geq 1$, $0 < p_0 < 1/2$, $r = p_0/(1-p_0)$ and $s = \min(r, 1-r)$. For the array (7.2),*

$$g(m,n;k) \geq (rs/8)^N \qquad \text{whenever } |m-n| \leq k \leq m+n \text{ and } m+n+k \text{ is even.}$$

All other coefficients vanish.

Proof. The vanishing statements are Lemma 2.1 and the symmetry used above. We use $\binom{N}{k} \leq 2^N$ throughout. Up to the order of m, n, there are three cases.

If m, n, k are even, the coefficient is a Krawtchouk coefficient computed in A, so Lemma 3.1 gives the lower bound $(s/4)^N \geq (rs/8)^N$.

Let $m = 2a$, $n = 2b+1$ and $k = 2c+1$, so that

$$\max(a-b-1,\, b-a) \leq c \leq \min(a+b,\, N-1).$$

Fix l and a set $U \not\ni l$ with $|U| = c$. The lth coordinate of $U_m U_n$ is

$$\binom{N}{a}^{-1}\binom{N-1}{b}^{-1} \sum_{|T|=a} \sum_{S \not\ni l,\ |S|=b} f_l\, y_{T\setminus\{l\}}\, y_S.$$

Every nonzero term arising from a chosen reduction of $y_{T\setminus\{l\}} y_S$ is a product of at most $N-1$ factors from $\{r, 1-r\}$, and therefore has coefficient at least s^{N-1}. Put $a' = |T \setminus \{l\}|$, which can equal $a-1$ when $a \geq 1$ and a when $a \leq N-1$. In the $N-1$ coordinates other than l, sets of sizes a' and b with overlap o, retaining the linear term on $t \leq o$ overlap coordinates, give monomials of size $a' + b - 2o + t$. As o ranges over its admissible values these sizes fill the interval $\big[|a'-b|, \min(a'+b, N-1)\big]$. The two admissible values of a' cover every c above. Hence $f_l y_U$ has coefficient at least $s^{N-1}\binom{N}{a}^{-1}\binom{N-1}{b}^{-1}$, and comparison with U_k gives

$$g(m,n;k) \geq \binom{N-1}{c} \frac{s^{N-1}}{\binom{N}{a}\binom{N-1}{b}} \geq \frac{2s^{N-1}}{4^N} \geq \left(\frac{s}{4}\right)^N.$$

Finally let $m = 2b+1$, $n = 2c+1$ and $k = 2a$. With π_i as in the proof of Lemma 2.1, (7.2) gives $\pi_{2a} = \binom{N}{a} r^{-a}$ and $\pi_{2c+1} = \binom{N-1}{c} r^{-c}/p_0$. Orthogonality gives

$$g(2b+1, 2c+1; 2a) = g(2a, 2b+1; 2c+1)\frac{\pi_{2a}}{\pi_{2c+1}} \geq \frac{2s^{N-1}}{4^N} \cdot \frac{p_0 r^N}{2^{N-1}} \geq \frac{2s^{N-1} r^{N+1}}{8^N},$$

using $r^{c-a} \geq r^N$ and $p_0 \geq r/2$. Since $s \leq r$, the last quantity is at least $(rs/8)^N$. ∎

The proof of Theorem 3.2 uses only three properties of its reference: nonnegative linearization with sums one, a lower bound for the upward coefficients, and a margin on the known support. We record it in that generality.

Lemma 7.5 *(Stability of a reference with a margin).* *Let a, b, c satisfy (2.1) on the indices $0, \dots, M$, have nonnegative linearization, and satisfy $a_i \geq \alpha$ for $i < M$, where $0 < \alpha \leq 1$. Suppose $g(m,n;k) \geq \delta > 0$ on a set Σ of index triples. Let $\widetilde{a}, \widetilde{b}, \widetilde{c}$ satisfy (2.1), with $\widetilde{b}_i \geq 0$, and differ from a, b, c by at most η in every entry. If $\eta \leq \alpha/2$ and $\eta(8/\alpha)^M \leq \delta/2$, then $|\widetilde{g}(m,n;k) - g(m,n;k)| \leq \delta/2$ for all m, n, k; in particular $\widetilde{g} \geq \delta/2$ on Σ.*

Proof. Repeat the proof of Theorem 3.2 with N replaced by M. Now $\widetilde{a}_m \geq \alpha/2$, so the subtracted recurrences give $e_{m+1} \leq (4e_m + 2e_{m-1} + 12\eta)/\alpha$. With $L = 8/\alpha \geq 8$, induction gives $e_m \leq \eta L^m$: the base case is $e_1 \leq 3\eta$, and the step needs $4L^m + 2L^{m-1} + 12 \leq 8L^m$, which holds for $m \geq 1$. Every entry of $\widetilde{B}_m - B_m$ is therefore at most $\eta L^M \leq \delta/2$, and (2.3) identifies these entries with the product coefficients. ∎

Theorem 3.2 is the case $M = N$, $\alpha = 1/N$ and $\delta = (s/4)^N$ of this lemma.

Theorem 7.6 *(Symmetric criterion near Szwarc's system).* *Let $N \geq 1$, $0 < p_0 < 1/2$, $r = p_0/(1-p_0)$, $s = \min(r, 1-r)$, and*

$$\eta = \frac{1}{2}\Big(\frac{rs}{8}\Big)^N (16N)^{-2N}.$$

Let c_k be the downward coefficients in (7.2). Take real numbers $\widetilde{c}_1, \dots, \widetilde{c}_{2N-1}$ with $|\widetilde{c}_k - c_k| \leq \eta$, and put $\widetilde{c}_0 = 0$, $\widetilde{c}_{2N} = 1$, $\widetilde{b}_k = 0$ and $\widetilde{a}_k = 1 - \widetilde{c}_k$. The resulting symmetric finite system has nonnegative linearization with sums one. Moreover $\widetilde{g}(m,n;k) \geq \frac{1}{2}(rs/8)^N$ whenever $|m-n| \leq k \leq m+n$ and $m+n+k$ is even, and $\widetilde{g}(m,n;k) = 0$ otherwise.

Proof. The perturbed array satisfies (2.1) with $M = 2N$: $\widetilde{a}_k \geq 1/N - \eta > 0$ for even $k < 2N$, $\widetilde{a}_k \geq 1/2 - \eta > 0$ for odd k, and $\widetilde{c}_k \geq \min(p_0, 1/N) - \eta > 0$ for $k > 0$. Theorem 7.1 shows that the reference has nonnegative linearization, and $a_k \geq 1/(2N)$ for $k < 2N$. Lemma 7.4 gives the margin $\delta = (rs/8)^N$ on the stated set. Apply Lemma 7.5 with $\alpha = 1/(2N)$: then $\eta \leq \alpha/2$ and $\eta(16N)^{2N} = \delta/2$. Off the stated set, $\widetilde{g}$ vanishes by Lemma 2.1 and by the symmetry of the perturbed measure, which holds because $\widetilde{b} \equiv 0$. The sums equal one because $\widetilde{R}_m(1) = 1$ and 1 lies in the support. ∎

Corollary 7.7 *(Symmetric systems outside both families).* *For $N \geq 2$, $0 < p_0 < 1/2$ and $0 < h \leq \eta$, the array obtained from (7.2) by $c_1 \mapsto p_0 + h$, $a_1 \mapsto 1 - p_0 - h$ has nonnegative linearization. It is neither of the form (7.2) nor of the form (4.2). More generally, Theorem 7.6 admits a relatively open set of symmetric stochastic arrays, with $2N-1$ free coefficients, around each array (7.2) with $0 < p_0 < 1/2$.*

Proof. Theorem 7.6 applies. The odd downward coefficients $c_1 = p_0 + h$ and $c_3 = p_0$ differ, so the array is not of the form (7.2); here $N \geq 2$ ensures $3 < 2N$. In (4.2), on $2N+1$ indices, the downward coefficients $c_i = r[i]_q/[2N]_q$ are strictly increasing, whereas $c_3 < c_1$. ∎

The quadratic decomposition [12, Theorem 4.1] also transfers the perturbation criteria: a symmetric array has nonnegative linearization whenever its two normalized components satisfy the hypotheses of Theorem 3.2, or those of Lemma 7.5 around a reference with a margin, provided the set Σ of that lemma contains every coefficient that can be nonzero for the perturbed components, or the remaining coefficients are separately known to stay nonnegative. Lemma 7.5 controls only Σ: for example, starting from $K_{2,1}$, keeping $b \equiv 0$ and $c_2 = 1$, and setting $c_1 = \frac{1}{2} + \varepsilon$, $a_1 = \frac{1}{2} - \varepsilon$ with $0 < \varepsilon < \frac{1}{2}$ gives $g(2,2;2) = -2\varepsilon/(\frac{1}{2} - \varepsilon) < 0$, although the coefficients positive at the reference remain positive. Conversely, the even

component of every array admitted by Theorem 7.6 is a nonsymmetric finite system with nonnegative linearization.

8. Inequality criteria away from the deep products

Szwarc's inequality method [20, Theorems 3.2 and 3.4] survives the passage to finite systems for all products whose two factors have total degree at most N. Only the terminal coefficients are excluded from its hypotheses. For the same products, Askey [1, Part II, Corollary 1] obtained nonnegativity under his own monotonicity hypotheses. Write a finite system in a positive renormalization $P_n = R_n/\sigma_n$ ($\sigma_0 = 1$) as $xP_n = \gamma_n P_{n+1} + \beta_n P_n + \alpha_n P_{n-1}$, so that $\gamma_n = a_n\sigma_{n+1}/\sigma_n$, $\beta_n = b_n$, $\alpha_n = c_n\sigma_{n-1}/\sigma_n$ and $\gamma_N = 0$.

Theorem 8.1 *(Low products).* *Let a, b, c satisfy (2.1). Suppose that for some positive σ,*

1. *β_n is nondecreasing for $0 \le n \le N-1$, and α_n is nondecreasing for $0 \le n \le N$;*
2. *$\alpha_n + \gamma_n$ is nondecreasing for $0 \le n \le N-2$;*
3. *$\gamma_i \ge \alpha_j$ whenever $0 \le j \le i$ and $i + j \le N-2$.*

Then $g(n, m; k) \ge 0$ whenever $n + m \le N$. Consequently a product coefficient can be negative only for a triple (n, m, k) in which each two indices have sum greater than N.

Proof. Coefficients in two positive normalizations differ by positive factors, so we may work with the polynomials $\widetilde{P}_n$ of [20, (3.9)], which satisfy $x\widetilde{P}_n = \alpha_{n+1}\widetilde{P}_{n+1} + \beta_n\widetilde{P}_n + \gamma_{n-1}\widetilde{P}_{n-1}$ with $\alpha_{N+1} = 0$. Fix k and let $u(n, m)$ be the coefficient of $\widetilde{P}_k$ in $\widetilde{P}_n\widetilde{P}_m$. Multiplying the product by x in the two ways shows $Hu = 0$ at every point of $\{0, \dots, N\}^2$, where $H = L_1 - L_2$ is the operator of [20, §3.4] built from these coefficients, and $u(n, 0) = \mathbf{1}_{n=k}$, $u(n, m) = u(m, n)$.

We prove $u(n, m) \ge 0$ for $n \ge m$, $n + m \le N$, by induction on m, following [20, Theorem 3.4]. Let v be the indicator of the points (i, j) with $j < m$, $|i - n| < m - j$ and $n + m - i - j$ odd. Every point in the support of v satisfies $j < i$ and $i + j \le n + m - 1 \le N - 1$, so $\langle Hu, v\rangle = 0$ involves only rows where the finite recurrence holds. Hence $0 = \langle u, H^*v\rangle$, and every point in the support of H^*v has $i + j \le n + m \le N$. With $\alpha_0 = 0$ and v set to zero outside its support, transposition gives

$$(H^*v)(i, j) = \alpha_i v(i-1, j) + \gamma_i v(i+1, j) + (\beta_i - \beta_j)v(i, j) - \alpha_j v(i, j-1) - \gamma_j v(i, j+1),$$

as in [20, §3.4]. At (n, m) it is $-\alpha_m < 0$. At the other points it is one of

$$\beta_i - \beta_j, \quad (\alpha_i + \gamma_i) - (\alpha_j + \gamma_j), \quad \alpha_i - \alpha_j, \quad \gamma_i - \alpha_j,$$

with $j \le i$. The first occurs only for $i \le N - 1$. The second occurs only at interior points, where $i + j \le n + m - 2 \le N - 2$. The third occurs on the edge $i + j = n + m$. The last occurs on the edge $i - j = n - m$, where $i + j \le N - 2$. Hypotheses (i)–(iii) make all of these nonnegative, and all such points have second coordinate less than m. By the induction hypothesis and $u(i, 0) \ge 0$, $\alpha_m u(n, m) = \sum_{(i,j)\ne(n,m)} (H^*v)(i, j)u(i, j) \ge 0$. For the last statement, $g(n, m; k)$, $g(n, k; m)$ and $g(m, k; n)$ are positive multiples of $\int R_n R_m R_k \, d\mu$. ∎

Corollary 8.2 *(Convex profiles).* *Let $b_i = (1-r)d_i$, $c_i = rd_i$, $a_i = 1 - d_i$ with $0 < r \le 1$ and a convex profile $0 = d_0 < d_1 < \cdots < d_N = 1$, that is, with nondecreasing increments. Then $g(n, m; k) \ge 0$ whenever two of n, m, k have sum at most N. This includes every array (4.2).*

Proof. Take $\sigma_n = r^{n/2}$. Then $\alpha_n = \sqrt{r}\, d_n$ and $\gamma_n = \sqrt{r}\,(1 - d_n)$, so α and $\beta = (1-r)d$ are nondecreasing and $\alpha_n + \gamma_n = \sqrt{r}$ is constant. Convexity and $d_0 = 0$ give $d_i + d_j \le d_{i+j}$, so $\gamma_i - \alpha_j = \sqrt{r}\,(1 - d_i - d_j) \ge \sqrt{r}\,(1 - d_{i+j}) > 0$ for $i + j \le N - 2$. Theorem 8.1 applies. ∎

For such profiles, positivity can therefore fail only on triples with all pairwise sums above N. The convex counterexamples recorded in Appendix A fail at (N, N, N), which is such a triple.

The cone argument extends to all products if the monotonicity conditions in (i)–(ii) are imposed through the terminal index N and the edge inequalities in (iii) are imposed for every $0 \le j \le i \le N - 1$.

Theorem 8.3 *(All products).* *Let a, b, c satisfy (2.1), and put $\gamma_N = 0$. Suppose that for some positive σ,*

1. *β_n and α_n are nondecreasing for $0 \le n \le N$;*
2. *$\alpha_n + \gamma_n$ is nondecreasing for $0 \le n \le N$;*
3. *$\gamma_i \ge \alpha_j$ whenever $0 \le j \le i \le N - 1$.*

Then the system has nonnegative linearization.

Proof. Use the notation of the proof of Theorem 8.1. Since $\alpha_{N+1} = 0$, the relation $x\widetilde{P}_N = \beta_N \widetilde{P}_N + \gamma_{N-1}\widetilde{P}_{N-1}$ is the terminal relation, which holds on the support. So $Hu = 0$ at every point of $\{0, \dots, N\}^2$, as noted there. We prove $u(n, m) \ge 0$ for all $n \ge m$ by induction on m. Let v be the indicator of the points (i, j) with $0 \le i \le N$, $j < m$, $|i - n| < m - j$ and $n + m - i - j$ odd. Then $0 = \langle Hu, v\rangle = \langle u, H^*v\rangle$, and H^*v is given by the formula in that proof. At $i = N$ the term $\gamma_N v(N + 1, j)$ is absent; this is the convention $\gamma_N = 0$. Hence $H^*v(n, m) = -\alpha_m$, and at every other point (i, j) with $H^*v(i, j) \ne 0$ we have $j < m$, $j \le i \le N$, and $H^*v(i, j)$ is one of

$$\beta_i - \beta_j, \quad (\alpha_i + \gamma_i) - (\alpha_j + \gamma_j), \quad \alpha_i - \alpha_j, \quad \gamma_i - \alpha_j.$$

The last one occurs only on the edge $i - j = n - m$, where $i \le n - 1 \le N - 1$. Hypotheses (i)–(iii) make all of these nonnegative. The induction hypothesis then gives $\alpha_m u(n, m) \ge 0$, and the case $n < m$ follows by symmetry. ∎

Corollary 8.4 *(Heavy top step).* *Let $b_i = (1-r)d_i$, $c_i = rd_i$, $a_i = 1 - d_i$ with $0 < r \le 1$ and an increasing profile $0 = d_0 < d_1 < \cdots < d_N = 1$ with $d_{N-1} \le \frac{1}{2}$. Then the system has nonnegative linearization.*

Proof. Take $\sigma_n = r^{n/2}$. Then $\alpha_n = \sqrt{r}\, d_n$ and $\gamma_n = \sqrt{r}\,(1 - d_n)$ for $n < N$, $\gamma_N = 0$, and $\beta_n = (1 - r)d_n$. So α and β are nondecreasing, and $\alpha_n + \gamma_n = \sqrt{r}$ for every $0 \le n \le N$, because $d_N = 1$. For $j \le i \le N - 1$, $\gamma_i - \alpha_j = \sqrt{r}\,(1 - d_i - d_j) \ge \sqrt{r}\,(1 - 2d_{N-1}) \ge 0$. Theorem 8.3 applies. ∎

The profile need not be convex. The corollary contains the arrays (4.2) with $q \ge 2$, since then $d_{N-1} = (q^{N-1} - 1)/(q^N - 1) \le \frac{1}{2}$, but not the Krawtchouk arrays with $N \ge 3$. The bound $d_{N-1} \le \frac{1}{2}$ cannot be relaxed in general: for $N = 2$ and $r = 1$, $g(2, 2; 2) = (1 - 2d_1)/(1 - d_1)$. For constant-ratio arrays with a convex profile, $d_i + d_j \le d_{i+j} \le 1$ whenever $i + j \le N$, so the only terms that can be negative in the proof are the edge terms $\gamma_i - \alpha_j = \sqrt{r}\,(1 - d_i - d_j)$ with $i + j > N$. For the Krawtchouk arrays these edge coefficients are $\sqrt{r}\,(N - i - j)/N$; any negative weighted contributions they make to the cone identity must be compensated by its other terms.

Deep triples and reversal. Call (n, m, k) deep if each two of its indices have sum greater than N. The reversed array $a'_i = c_{N-i}$, $b'_i = b_{N-i}$, $c'_i = a_{N-i}$ need not satisfy $a'_0 = 1$, $b'_0 = 0$. Its polynomials R'_i are defined by the reversed recurrence from $R'_0 = 1$, so $R'_1 = (x - b_N)/c_N$, and they still satisfy $R'_i(1) = 1$; its matrix is $\Pi J \Pi$, where Π reverses the order of the indices. Write g' for its product coefficients.

Lemma 8.5 *(Reversal identity). On the support, $R_N R'_t = R_{N-t}$ for $0 \le t \le N$.*

Proof. From $e_0^{\mathsf{T}} R_i(J) = e_i^{\mathsf{T}}$ for the array and for its reversal, $e_N^{\mathsf{T}} R'_t(J) = e_{N-t}^{\mathsf{T}}$. Hence $e_0^{\mathsf{T}} R_N(J) R'_t(J) = e_N^{\mathsf{T}} R'_t(J) = e_0^{\mathsf{T}} R_{N-t}(J)$. Both matrices commute with J and e_0 is cyclic, so $R_N(J) R'_t(J) = R_{N-t}(J)$, which is the claim on the spectrum. ∎

The case $t = N$, $R_N R'_N = 1$ on the support, is Lemma 2 of de Boor and Golub [3], stated there for monic polynomials of a symmetric Jacobi matrix and its reversal.

Theorem 8.6 *(Deep reduction). For every array satisfying (2.1) and every deep triple (x, y, z),*

$$g(x, y; z) = \sum_{t=0}^{N} g'(N - x, N - y; t)\, g(N, N - t; z).$$

Moreover $g(N, N; N - l)$ is the coefficient of R'_l in R_N. Hence all coefficients are nonnegative if (a) $g \ge 0$ on non-deep triples, (b) $g'(i, j; t) \ge 0$ whenever $i + j < N$, and (c) R_N is a nonnegative combination of $R'_0, \dots, R'_N$.

Proof. By Lemma 8.5, $R_x = R_N R'_{N-x}$ and $R_y = R_N R'_{N-y}$ on the support. Expanding $R'_{N-x} R'_{N-y} = \sum_t g'(N - x, N - y; t) R'_t$ and using Lemma 8.5 again gives $R_x R_y = \sum_t g'(N - x, N - y; t) R_N R_{N-t}$. Expanding $R_N R_{N-t}$ gives the formula. Writing $R_N = \sum_l e_l R'_l$, Lemma 8.5 gives $R_N^2 = \sum_l e_l R_{N-l}$, so $e_l = g(N, N; N - l)$. For the last statement, $(N - x) + (N - y) < N$, so (b) makes the weights g' nonnegative. It remains to see $g(N, n; z) \ge 0$. If (N, n, z) is not deep this is (a). Otherwise $n + z > N$, and $g(N, n; z)$ is a positive multiple of $g(n, z; N) = \sum_t g'(N - n, N - z; t)\, g(N, N - t; N)$; here $g(N, N - t; N)$ is a positive multiple of $g(N, N; N - t)$, which is nonnegative by (c). ∎

Theorem 8.7 *(Palindromic criterion).* *Let a, b, c satisfy (2.1) with $a_n = c_{N-n}$ and $b_n = b_{N-n}$ for all n, and satisfy the hypotheses of Theorem 8.1 for some positive σ. Then the system has nonnegative linearization.*

Proof. The reversed array equals the array, so $R' = R$ and $g' = g$. Theorem 8.1 gives $g \geq 0$ on all non-deep triples, in particular $g'(i, j; t) \geq 0$ for $i + j < N$. Since $R_N = R'_N$ is a single reversed polynomial, $g(N, N; \cdot) \geq 0$. Theorem 8.6 gives the deep triples. ∎

Corollary 8.8 *(Monotone palindromic systems).* *Let $b \equiv 0$, $a_n = 1 - c_n$, and let $0 = c_0 < c_1 \leq c_2 \leq \cdots \leq c_N = 1$ satisfy $c_n + c_{N-n} = 1$. Then the system has nonnegative linearization. This includes the symmetric Krawtchouk array $c_n = n/N$.*

Proof. The array is palindromic. With $\sigma \equiv 1$, $\alpha_n = c_n$ is nondecreasing, $\beta \equiv 0$ and $\alpha_n + \gamma_n = 1$. If $j \leq i$ and $i + j \leq N - 2$, then $j < N - i$, so $\alpha_j = c_j \leq c_{N-i} = 1 - c_i = \gamma_i$. Theorem 8.7 applies. ∎

Within the palindromic subclass $b \equiv 0$, $c_n + c_{N-n} = 1$, the hypotheses of Corollary 8.8 are monotonicity inequalities. They contain the symmetric Krawtchouk endpoint, where the perturbation criterion of Section 3 gives no neighbourhood; away from it, the convex counterexamples of Appendix A show that convexity alone does not suffice.

Combined with the quadratic decomposition [12, Theorem 4.1], Corollary 8.8 gives an inequality criterion for symmetric systems: an array with $b \equiv 0$ on $0, \ldots, 2N$ has nonnegative linearization whenever its two normalized components satisfy the hypotheses of Corollary 8.8. For Szwarc's system (7.2) at $p_0 = 1/2$ the components are the symmetric Krawtchouk arrays on $N + 1$ and N points [12, Corollary 4.2], so this case is covered by monotonicity inequalities within the palindromic subclass.

The reversal route cannot reach systems whose diagonal increases at the top once $N \geq 3$. If $b_{N-1} < b_N$, the reversed array has $g'(1, 1; 1) = (b'_1 - b'_0)/a'_0 = (b_{N-1} - b_N)/c_N < 0$, and condition (b) of Theorem 8.6 requires $g'(1, 1; 1) \geq 0$ when $1 + 1 < N$. In particular it fails for the Krawtchouk arrays with $r < 1$ and $N \geq 3$. Szwarc's array (7.2) has $b \equiv 0$, so this argument does not apply to it. Its reversal replaces p_0 by $1 - p_0$ and gives $g'(2, 2; 2) = (2p_0 - 1)/(Np_0) < 0$ for $p_0 < 1/2$, which violates condition (b) for the array of length $2N + 1$ when $4 < 2N$, that is, $N \geq 3$. Miclo's palindromic criterion [10, Theorem 31] requires $2^n P(n, n + 1)$ to be nonincreasing on $[0, N/2]$, which the symmetric Krawtchouk chain violates for $N > 2$.

9. Limits of inequality criteria

The radius of Theorem 3.2 shrinks like $(1 - r)^N$ as $r \to 1$ and vanishes at the symmetric endpoint, and Theorem 7.6 needs $p_0 < 1/2$. This section shows that this degeneration is forced. At the symmetric Krawtchouk systems the corner coefficient $g(N, N; N)$ vanishes and has a nonzero first-order change along most directions (Lemma 9.2); this yields the two equalities for uniform deformation rules (Theorem 9.4) and the exact order $(1 - r)^N$ of the stability radius (Corollary 9.11), which Theorem 3.2 attains up to a factor depending on N.

Theorems 8.1 and 8.7 control every triple except the deep ones by inequalities, the criterion of [12, Theorem 3.1] controls all triples of constant-ratio arrays with log-concave Newton profiles, and Appendix A records shape conditions that fail on deep triples. This section shows that part of the failure is structural near the symmetric endpoint. For uniformly sampled polynomial deformation directions, and for the analytic and boundary-layer extensions specified below, admission at every sufficiently large symmetric Krawtchouk system imposes two first-order boundary equalities. This does not exclude arbitrary uniform inequality criteria, nor families of admissible arrays that are not of these forms. Write $K_{N,r}$ for the Krawtchouk array (4.2) with $q = 1$, and use the coordinates $b_1, \dots, b_{N-1}, c_1, \dots, c_N$ of a normalized array with $b_0 = 0$, $a_0 = 1$, $a_N = 0$, $b_N = 1 - c_N$ and $a_n = 1 - b_n - c_n$ otherwise. An affine change of x, together with the positive normalization $R_n(x_0) = 1$ at the largest support point x_0, reduces every finite system to this form. A direction $v = (\delta b_1, \dots, \delta b_{N-1}, \delta c_1, \dots, \delta c_N)$ changes these coordinates; we put $\delta b_0 = \delta c_0 = 0$ and $\delta b_N = -\delta c_N$.

Lemma 9.1 *(Corner value).* *For $0 < r \le 1$, the array $K_{N,r}$ has $g(N, N; N) = (1 - r)^N$.*

Proof. In the model of Lemma 3.1, $U_N = y_1 \cdots y_N$ is the only normalized basis element containing $y_{\{1,\dots,N\}}$, and $U_N^2 = \prod_i \big((1 - r)y_i + r\big)$ has coefficient $(1 - r)^N$ at that monomial. ∎

Thus the corner coefficient vanishes at $r = 1$, where it is a structural zero, together with the other perimeter and parity zeros of the palindromic array. The next lemma computes its first-order change exactly.

Lemma 9.2 *(Corner identity).* *Let $N \ge 2$ and $s_N = (-1)^{\lfloor N/2 \rfloor + 1} \binom{N-1}{\lfloor (N-1)/2 \rfloor}^{-1}$. Put $F = \delta b + 2\delta c$ if N is even and $F = \delta b$ if N is odd. Then*

$$\frac{d}{d\varepsilon}\, g(N, N; N)\Big|_{K_{N,1} + \varepsilon v,\ \varepsilon = 0} = s_N \sum_{n=0}^{N} (-1)^n \binom{N}{n} F(n).$$

Proof. At $K_{N,1}$ the nodes are $\lambda_j = 1 - 2j/N$, with masses $w_j = 2^{-N}\binom{N}{j}$ and $\pi_n = \binom{N}{n}$ in Lemma 2.1. Moreover $\binom{N}{n} R_n(\lambda_j)$ is the binary Krawtchouk polynomial, so that

$$\sum_{j=0}^{N} w_j R_n(\lambda_j) s^j = 2^{-N}(1 + s)^{N-n}(1 - s)^n. \tag{9.1}$$

The vectors $u_j = (R_n(\lambda_j))_n$ and $v_j = (w_j \pi_n R_n(\lambda_j))_n$ are right and left eigenvectors of J with $v_i^{\mathsf{T}} u_j = \mathbf{1}_{i=j}$. Since $R_N(\lambda_j) = (-1)^j$, the matrix $\Pi = R_N(J)$ reverses the indices. Let $J_\varepsilon = J + \varepsilon E$ be the perturbed recurrence matrix and R_N^ε the perturbed polynomial. The matrix $R_N^\varepsilon(J_\varepsilon)$ commutes with J_ε and has first row e_N^{T}, by (2.3). Its derivative X at $\varepsilon = 0$ therefore satisfies $[J, X] = [\Pi, E]$ and $e_0^{\mathsf{T}} X = 0$. In the eigenbasis, $\widehat{X}_{ij} = v_i^{\mathsf{T}} X u_j$ equals $((-1)^i - (-1)^j)\widehat{E}_{ij}/(\lambda_i - \lambda_j)$ for $i \ne j$, where $\widehat{E}_{ij} = v_i^{\mathsf{T}} E u_j$, and $\sum_i \widehat{X}_{ij} = 0$. By (2.3), and since $\pi_N = 1$, the derivative in the lemma is $h_N := e_N^{\mathsf{T}} X e_N = \sum_{i,j} (-1)^{i+j} \widehat{X}_{ij} w_j$.

Eliminating $\widehat{X}_{jj}$ gives

$$h_N \;=\; -2\sum_{i+j\ \text{odd}} \widehat{X}_{ij}w_j \;=\; N\sum_{i\neq j}\widehat{E}_{ij}w_j\,\frac{(-1)^i-(-1)^j}{i-j}.$$

Perturbing row n alone means $E = e_n\rho^{\mathsf{T}}$ and $\widehat{E}_{ij} = w_i\pi_n R_n(\lambda_i)\,\rho^{\mathsf{T}}u_j$. Write i for the imaginary unit and insert

$$\frac{(-1)^i-(-1)^j}{i-j} \;=\; -\pi\int_0^1 \sin\pi\big(\theta i+(1-\theta)j\big)\,d\theta,$$

which also vanishes for $i = j$. Then

$$h_N \;=\; -\pi N\pi_n\int_0^1 \operatorname{Im}\big(A(\theta)B(\theta)\big)\,d\theta,$$
$$A \;=\; \sum_i w_iR_n(\lambda_i)e^{\mathrm{i}\pi\theta i}, \qquad B \;=\; \sum_j w_j(\rho^{\mathsf{T}}u_j)e^{\mathrm{i}\pi(1-\theta)j},$$

and both sums are given by (9.1). Put $\phi = \pi\theta/2$.

- Coordinate b_n, $1 \le n < N$, so $\rho = e_n - e_{n+1}$. Then

  $$AB \;=\; -(-1)^n\,\mathrm{i}^{N-1}\cos^N\phi\,\sin^{N-1}\phi\,e^{-\mathrm{i}\phi},$$

  and $h_N = s_N(-1)^n\binom{N}{n}$. The integral reduces to $\int_0^{\pi/2}\cos^{N+1}\phi\,\sin^{N-1}\phi\,d\phi$ (N even) or $\int_0^{\pi/2}\cos^N\phi\,\sin^N\phi\,d\phi$ (N odd), and each equals $|s_N|/(2N)$.

- Coordinate c_n, $1 \le n < N$, so $\rho = e_{n-1} - e_{n+1}$. Now B acquires the factor $e^{\mathrm{i}\phi}/\cos\phi$, and the integrand carries $\sin((N-1)\pi/2)$. The contribution vanishes for odd N. For even N it is twice that of b_n, since the two beta integrals have ratio 2.

- Coordinate c_N, so $\rho = e_{N-1} - e_N$ in row N. The same computation with $n = N$ gives s_N.

Summing these contributions with the conventions $\delta b_0 = \delta c_0 = 0$ and $\delta b_N = -\delta c_N$ gives the stated alternating sum. ∎

Definition 9.3. A *uniform rule* is a pair of real functions β, γ on $[0,1]$. It defines, for each N, the direction v_N with $\delta b_n = \beta(n/N)$ for $1 \le n < N$ and $\delta c_n = \gamma(n/N)$ for $1 \le n \le N$. A set of arrays *admits* the rule at the symmetric endpoint if there is N_0 such that for every $N \ge N_0$ there is $\varepsilon_N > 0$ with $K_{N,1} + \varepsilon v_N$ in the set for all $0 < \varepsilon < \varepsilon_N$. Put

$$L_{\text{even}} \;=\; \beta(0) + 2\gamma(0) + \beta(1) + \gamma(1), \qquad L_{\text{odd}} \;=\; \beta(0) - \beta(1) - \gamma(1).$$

Theorem 9.4 *(Forced equalities for uniform criteria).* *Let β, γ be polynomials. If a set of arrays with property (P) admits the rule (β, γ) at the symmetric endpoint, then $L_{\text{even}} = L_{\text{odd}} = 0$. If $L_{\text{even}} = L_{\text{odd}} = 0$, the first-order change of $g(N, N; N)$ along v_N vanishes for every N above the degrees of β and γ.*

Proof. For a polynomial p of degree less than N, $\sum_{n=0}^{N}(-1)^n\binom{N}{n}p(n/N) = 0$. Only boundary terms survive in Lemma 9.2: the derivative equals $-s_N L_{\mathrm{even}}$ for even N and $-s_N L_{\mathrm{odd}}$ for odd N. The sign of s_N changes from N to $N+2$. If, for instance, $L_{\mathrm{even}} \neq 0$, then the derivative is negative for every large even N in one residue class modulo 4. For such N, $g(N,N;N) < 0$ at $K_{N,1} + \varepsilon v_N$ for all small $\varepsilon > 0$, since the coefficient vanishes at $\varepsilon = 0$. ∎

The known positive deformations satisfy both equalities: the Krawtchouk direction $\beta(x) = x$, $\gamma(x) = -x$, palindromic deformations with $\beta = 0$, γ odd about $\frac{1}{2}$ and $\gamma(0) = \gamma(1) = 0$ (Corollary 8.8), and the tangent direction $\beta = 0$, $\gamma(x) = x(x-1)$ of the geometric family at $q = 1$ (Theorem 4.1). The two functionals are independent already on constant rules, where $L_{\mathrm{even}} = 2\beta + 3\gamma$ and $L_{\mathrm{odd}} = -\gamma$. Consequently, for every $d \geq 0$, no set of arrays with property (P) admits all rules of a nonempty open subset of the polynomial rules of degree at most d. In this sense, a criterion that admits the symmetric Krawtchouk systems through polynomial uniform rules must impose the two equalities on them. The geometric criterion and Corollary 8.8 are of this mixed type.

The polynomial hypothesis can be replaced by analyticity on a fixed region. Let $U(x) = \int_0^1 \log|x-t|\,dt$ and $\Omega = \{x \in \mathbb{C} : U(x) \leq -1\}$, the region of convergence of equispaced interpolation. It contains $[0,1]$, has nonanalytic boundary points (with vertical tangents) at 0 and 1, and meets the line $\operatorname{Re} x = \frac{1}{2}$ at $\operatorname{Im} x = \pm 0.26276\ldots$.

Lemma 9.5 *(Nörlund–Rice bound).* *If f is analytic on an open set containing Ω, then $\sum_{n=0}^{N}(-1)^n\binom{N}{n}f(n/N) = O(\sqrt{N}\,e^{-\eta N})$ for some $\eta > 0$.*

Proof. The residues of $1/\prod_{k=0}^{N}(x-k/N)$ give

$$\sum_{n=0}^{N}(-1)^n\binom{N}{n}f(n/N) = (-1)^N\frac{N!}{N^N}\,\frac{1}{2\pi i}\oint_\Gamma \frac{f(x)\,dx}{\prod_{k=0}^{N}(x-k/N)}$$

for any cycle Γ in the domain of f that winds once around $[0,1]$. Since U is continuous and tends to ∞, the compact sets $\{U \leq -1+t\}$ shrink to Ω as $t \downarrow 0$. Choose $\eta > 0$ with $\{U \leq -1+3\eta\}$ inside the domain of f, and a bounded polygonal open set V with $\{U \leq -1+\eta\} \subset V$ and $\overline{V} \subset \{U < -1+3\eta\}$. The oriented boundary Γ of V, possibly with several components, is a cycle outside $\{U \leq -1+\eta\}$ that winds once around every node k/N and bounds a region inside the domain of f. With $d = \operatorname{dist}(\Gamma,[0,1]) > 0$, the derivative of $t \mapsto \log|x-t|$ on $[0,1]$ is bounded by $1/d$ for $x \in \Gamma$, so a Riemann-sum estimate gives $\sum_{k=0}^{N}\log|x-k/N| \geq NU(x) - C_\Gamma \geq N(-1+\eta) - C_\Gamma$ with C_Γ independent of N. Stirling's bound $N!/N^N \leq e\sqrt{N}e^{-N}$ completes the proof. ∎

Corollary 9.6. *Let β,γ be analytic on an open set containing Ω. If a set of arrays with property (P) admits the rule (β,γ) at the symmetric endpoint, then $L_{\mathrm{even}} = L_{\mathrm{odd}} = 0$. When these equalities hold, the first-order change of $g(N,N;N)$ along v_N is $O(|s_N|\sqrt{N}\,e^{-\eta N})$; it need not vanish.*

Proof. By Lemma 9.5, the alternating sum of $\beta + 2\gamma$ or β is $O(\sqrt{N}e^{-\eta N})$. The derivative in Lemma 9.2 is therefore $s_N(-L_{\mathrm{even}} + o(1))$ for even N and $s_N(-L_{\mathrm{odd}} + o(1))$ for odd N,

and the argument of Theorem 9.4 gives the equalities. The rule $\beta(x) = e^x - 1 - (e-1)x$, $\gamma = 0$ has $L_{\mathrm{even}} = L_{\mathrm{odd}} = 0$ and derivative $s_N(1-e^{1/N})^N \neq 0$. ∎

Deformations may also depend on N in other ways. For each fixed N the symmetric endpoint is a limit of systems whose product coefficients are strictly positive on the whole triangle; the corner nevertheless constrains families that are regular in N. The first statement rests on the following lemma, valid for every finite system. Write μ for the orthogonality measure and x_0 for its largest support point, so that $R_n(x_0) = 1$.

Lemma 9.7 *(Lowering the top mass).* *Let $0 \le \varepsilon < \mu(\{x_0\})$, let $\mu_\varepsilon = \mu - \varepsilon\delta_{x_0}$, and let R_n^ε be its polynomials, normalized at x_0. Put $\pi_n^\varepsilon = \big(\int (R_n^\varepsilon)^2\, d\mu_\varepsilon\big)^{-1}$ and $\pi_n = \pi_n^0$; these inverse norms are not rescaled to impose $\pi_0^\varepsilon = 1$. The recurrence coefficients in (1) are taken in the fixed coordinate x, without an ε-dependent affine normalization.*

1. *The recurrence coefficients change at rate $(\dot a_n, \dot b_n, \dot c_n) = \pi_n(a_n,\, c_n - a_n,\, -c_n)$.*
2. *If $\int R_x R_y R_z\, d\mu = 0$, then*

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \int R_x^\varepsilon R_y^\varepsilon R_z^\varepsilon\, d\mu_\varepsilon = 2 - T_x(y,z) - T_y(x,z) - T_z(x,y),$$
$$T_x(y,z) = \sum_{k \ge x} g(y,z;k).$$

Proof. Write $R_n^\varepsilon = R_n + \varepsilon Q_n + O(\varepsilon^2)$, with $\deg Q_n \le n$ and $Q_n(x_0) = 0$. First-order orthogonality gives $\int Q_n p\, d\mu = p(x_0)$ for $\deg p < n$, hence $Q_n = \sum_{k<n} \pi_k (R_k - R_n)$, and $\pi_n^\varepsilon = \pi_n\big(1 + \varepsilon(2S_n + \pi_n)\big) + O(\varepsilon^2)$ with $S_n = \sum_{k<n} \pi_k$. For (1), differentiate $c_n = \pi_{n-1} \int x R_n R_{n-1}\, d\mu$ and $b_n = \pi_n \int x R_n^2\, d\mu$, using the recurrence, the row sums $a_m + b_m + c_m = x_0$ and $\pi_{n-1} a_{n-1} = \pi_n c_n$; the terms with S_n cancel, and $\dot a_n$ follows from the row sums. For (2), the derivative equals $-R_x R_y R_z(x_0) + \int (Q_x R_y R_z + R_x Q_y R_z + R_x R_y Q_z)\, d\mu$, and $\int Q_x R_y R_z\, d\mu = \sum_{k<x} g(y,z;k) = 1 - T_x(y,z)$. ∎

Proposition 9.8 *(Interior points near the symmetric endpoint).* *For every $N \ge 2$ there is $\Lambda_N < \infty$ such that, for $\Lambda > \Lambda_N$ and all sufficiently small $\varepsilon > 0$, the measure of $K_{N,1-\Lambda\varepsilon}$ with its top mass lowered by ε has $g(m,n;k) > 0$ for all $|m-n| \le k \le \min(m+n, N)$. Consequently, after the affine normalization of Section 2, a relatively open set of arrays near it has property (P).*

Proof. By the model of Lemma 3.1, $K_{N,r}$ has

$$g(m,n;k) = \sum_j H(m,n;j) \binom{j}{s} r^{j-s}(1-r)^s,$$
$$H(m,n;j) = \frac{\binom{m}{j}\binom{N-m}{n-j}}{\binom{N}{n}}, \qquad s = k - m - n + 2j,$$

where the sum is over $\max(0, m+n-N) \le j \le \min(m,n)$ with $0 \le s \le j$. At $r = 1$ only $s = 0$ survives, so a triangle coefficient of $K_{N,1}$ vanishes exactly when $m + n + k$ is

odd or $m+n+k > 2N$. Both deformations are smooth in (r,ε), so only these zeros need first-order positivity.

- If $m+n+k > 2N$, all three tails in Lemma 9.7(2) vanish, since $g(y,z;k) = 0$ for $k > 2N-y-z$. The top-mass rate is therefore $2\pi_k > 0$, and the rate of decreasing r is nonnegative because $g \geq 0$ for $r \leq 1$.
- If $m+n+k$ is odd and at most $2N+1$, the rate of decreasing r is the $s=1$ term $H(m,n;j)\,j$ with $j = (1+m+n-k)/2$. It is positive, because $j \geq 1$ and $\max(0, m+n-N) \leq j \leq \min(m,n)$.

These cases cover all zeros. Let $\mathcal{Z}_N$ be the set of triangle triples whose coefficient vanishes at $K_{N,1}$. For $z \in \mathcal{Z}_N$, write A_z for the top-mass rate and B_z for the rate of decreasing r, and define

$$\Lambda_N = \max\Big(\{0\} \cup \Big\{-\frac{A_z}{B_z} : z \in \mathcal{Z}_N,\ A_z < 0\Big\}\Big).$$

Every denominator appearing here is positive by the preceding cases. Hence $\Lambda > \Lambda_N$ makes every reference-zero derivative strictly positive, and finiteness of $\mathcal{Z}_N$ gives a common threshold for ε.

Here the measure of $K_{N,r}$ is its probability orthogonality measure. Choose ε so small that $r = 1-\Lambda\varepsilon > 0$ and $\varepsilon < (r/(1+r))^N$, so that the lowered top mass remains positive. The affine coordinate $y = (1-\varepsilon)x + \varepsilon = R_1^{\varepsilon}(x)$ restores the basic recurrence normalization and leaves all product coefficients unchanged. Strict positivity on the triangle gives $b_n = g(1,n;n) > 0$ for $1 \leq n \leq N$. Lemma 7.5, with Σ the whole triangle, therefore gives a relatively open neighborhood with property (P) in the affine space defined by the basic boundary and row-sum equalities. ■

Exact rational computation of the maximum above gives $\Lambda_N = 4, 10, 18, 28, 52, 90, 165$ for $N = 4, \ldots, 10$. These finite-size values do not establish an asymptotic growth law or a geometric thickness estimate. At $K_{N,1}$, in the fixed spectral coordinate, Lemma 9.7(1) gives

$$\dot{c}_n = -\frac{n}{N}\binom{N}{n}, \qquad \dot{b}_n = \Big(\frac{2n}{N} - 1\Big)\binom{N}{n}.$$

After the affine normalization restoring $R_1 = x$, the top-mass direction is

$$\delta a_n = \frac{N-n}{N}\Big(\binom{N}{n} - 1\Big), \qquad \delta b_n = 1 + \Big(\frac{2n}{N} - 1\Big)\binom{N}{n}, \qquad \delta c_n = -\frac{n}{N}\Big(1 + \binom{N}{n}\Big).$$

Its binomial dependence on N is not a uniform rule of Definition 9.3, consistently with Theorem 9.4.

Proposition 9.9 *(Families depending on N).* *Let v_N be directions such that a set with property (P) contains $K_{N,1} + \varepsilon v_N$ for all large N and small $\varepsilon > 0$, and let A_N be the alternating sum in Lemma 9.2 for v_N. Then $s_N A_N \geq 0$ for all large N.*

1. *Along either parity class, A_N is not eventually strictly positive and not eventually strictly negative. If A_N has an eventual weak sign along a parity class, it vanishes for*

all large N of the residue class modulo 4 on which that sign conflicts with the sign of s_N.

2. *Suppose $\delta b_n = \beta(n/N) + p_n + q_{N-n}$ and $\delta c_n = \gamma(n/N) + p'_n + q'_{N-n}$, with β, γ as in Corollary 9.6 and finitely supported boundary layers p, q, p', q' independent of N. Then $q'_0 = L_{\mathrm{even}} = L_{\mathrm{odd}}$, $p_k = q_k$ and $p'_k + q'_k = -p_k$ for $k \geq 1$.*

Proof. A negative first-order change of $g(N, N; N)$ excludes $K_{N,1} + \varepsilon v_N$ for small ε, so $s_N A_N \geq 0$. Part (1) follows because the sign of s_N alternates along each parity class. In (2), write $D_N f = \sum_{n=0}^{N} (-1)^n \binom{N}{n} f(n/N)$. For large N, $A_N = D_N(\beta + 2\gamma) + P_e(N)$ for even N and $A_N = D_N \beta + P_o(N)$ for odd N, where

$$P_e(N) = q'_0 - L_{\mathrm{even}} + \sum_{k \geq 1} (-1)^k \binom{N}{k} \left[p_k + q_k + 2(p'_k + q'_k) \right],$$

$$P_o(N) = q'_0 - L_{\mathrm{odd}} + \sum_{k \geq 1} (-1)^k \binom{N}{k} (p_k - q_k).$$

These are polynomials in N. A nonzero polynomial has an eventual strict sign, which the remainder $O(\sqrt{N} e^{-\eta N})$ of Lemma 9.5 cannot change, so part (1) forces $P_e \equiv P_o \equiv 0$. The polynomials $1, \binom{N}{1}, \binom{N}{2}, \ldots$ are linearly independent, which gives the stated conditions. ∎

Thus a separate adjustment of the top coefficient c_N reduces the two bulk equalities to the single equality $\gamma(0) + \beta(1) + \gamma(1) = 0$. For $k \geq 1$ and $N \geq k + K + 1$, where K bounds the supports of p, q, p', q', the boundary-layer contributions satisfy the palindromic first-order relations $\delta b_k = \delta b_{N-k}$ and $\delta a_k = \delta c_{N-k}$; the separate terminal adjustment need not itself be palindromic. Families not covered by part (1) must have A_N change sign, or vanish, infinitely often within a parity class.

Criteria that cover $r < 1$ but not $r = 1$ are thin near $r = 1$.

Proposition 9.10 *(Thinness near the symmetric endpoint).* *Fix $N \geq 2$ and a direction v along which the first-order change $\dot{g}$ of $g(N, N; N)$ at $K_{N,1}$ is negative. There are $r_* < 1$ and $C, c > 0$ such that $K_{N,r} + \varepsilon v$ lacks property (P) whenever $r_* < r < 1$ and $C(1-r)^N \leq \varepsilon \leq c$.*

Proof. The function $G(r, \varepsilon) = g(N, N; N)$ at $K_{N,r} + \varepsilon v$ is a rational function of the recurrence entries whose denominators involve only upward coefficients, which are positive at $K_{N,1}$; so G is smooth near $(1, 0)$. By Lemma 9.1, $G(r, 0) = (1-r)^N$. Put $d = |\dot{g}|$. On a small rectangle around $(1, 0)$, on which the basic sign conditions persist, $\partial_\varepsilon G \leq -d/2$, hence $G(r, \varepsilon) \leq (1-r)^N - d\varepsilon/2 < 0$ for $\varepsilon \geq 4(1-r)^N/d$. ∎

Corollary 9.11 *(Order of the stability radius as $r \to 1$).* *For $N \geq 2$ let $\rho_N(r)$ be the largest ρ such that every array satisfying (2.1) with $\widetilde{b}_i \geq 0$ and with left side of (3.3) at most ρ has property (P). Then $\rho_N(r)$ is of exact order $(1-r)^N$ as $r \to 1$.*

Proof. Theorem 3.2 gives $\rho_N(r) \geq \eta_N(r) = \frac{1}{2}((1-r)/(32N))^N$ for $r \geq \frac{1}{2}$. For the upper bound, use the terminal direction $\delta c_N = -\operatorname{sgn}(s_N)$, $\delta b_N = \operatorname{sgn}(s_N)$, with all other entries

unchanged. By Lemma 9.2 its first-order change is $-|s_N| < 0$, and it moves the full array by exactly ε in the norm of (3.3). Put $t = 1 - r$ and $\varepsilon = Ct^N$ as in Proposition 9.10. For $N \geq 2$ and r near 1, $\varepsilon \leq t/2$ and $\varepsilon \leq r/2$, so $\widetilde{b}_N \geq t/2$ and $\widetilde{c}_N \geq r/2$; every other coefficient is unchanged. The array satisfies the basic conditions and $\widetilde{b}_i \geq 0$, lies at full-array distance $\varepsilon = C(1-r)^N$ from $K_{N,r}$, and lacks property (P). Hence $\rho_N(r) \leq C(1-r)^N$. The restriction $N \geq 2$ is needed: every admissible two-point array with $b_1 \geq 0$ has property (P). ■

For fixed r the radius also decays at least exponentially in N: raising the top mass of the measure of $K_{N,r}$, the reverse of Lemma 9.7, gives $\rho_N(r) \leq 3(1-r^2)^N$ whenever $(1-r^2)^N \leq \frac{1}{2}$ and $3(1-r^2)^N \leq (1-r)/N$ [12, Proposition A.2 and Corollary A.3].

Thus, for every fixed $r \in (0,1)$, no set of arrays with property (P) contains coefficient balls of a positive radius independent of N around all $K_{N,r}$; Theorem 3.2 gives the lower bound $\rho_N(r) \geq \frac{1}{2}(\min(r, 1-r)/(32N))^N$, which decays faster than every exponential. The deformation that raises the top mass has the binomial profile of the masses and is not a uniform rule. Along the terminal row, $g(N, N; N)$ is a polynomial in the perturbation, since the recurrence for the linearization coefficients divides only by $a_0, \ldots, a_{N-1}$. Exact failures of property (P) along the terminal row at fixed r, a heuristic for their first-order term, and exact counterexamples to convex-profile conditions are collected in Appendix A; they establish failures at specified parameters and sizes, without an asymptotic claim.

10. Open problems

Sharper stability radii. For fixed N and $0 < r < 1$, determine or estimate the largest coefficient radius for which every admissible stochastic perturbation has property (P). Corollary 9.11 shows that, for fixed $N \geq 2$, $\rho_N(r)$ has exact order $(1-r)^N$ as $r \to 1$, which the bound of Theorem 3.2 attains up to a constant depending on N, and [12, Corollary A.3] gives $\rho_N(r) = O((1-r^2)^N)$ for each fixed r; the gap to the lower bound $\frac{1}{2}(\min(r, 1-r)/(32N))^N$ of Theorem 3.2 remains open. The bound in Theorem 3.2 comes from a single-term lower margin and a worst-case recurrence amplification. Both estimates discard substantial information. A useful improvement would retain a transparent coefficient test and describe its dependence on both N and the bias parameter.

The symmetric endpoint. At $r = 1$, the reference margin in Lemma 3.1 vanishes because parity forces many coefficients to be zero. Symmetric perturbations preserve the parity zeros, and Theorem 7.6 exploits this for Szwarc's quadratic system when $p_0 < 1/2$. At $p_0 = 1/2$, and for the symmetric Krawtchouk array, further coefficients on the parity support vanish. Determine perturbation directions and structural constraints that preserve positivity there; continuity alone does not control coefficients emerging from those zeros. For every $N \geq 1$, the symmetric Krawtchouk array is a limit of arrays whose product coefficients are strictly positive throughout the triangle support. For $N \geq 2$ this follows from Proposition 9.8; the two-point case is immediate. Theorem 9.4 and Corollary 9.6 impose two boundary equalities on admitted polynomial rules and on the analytic rules satisfying

the holomorphic-extension hypothesis specified there. A description of the cone for all N is open.

Monotonicity-type criteria for finite systems. Szwarc's Theorems 3.2 and 3.3 in [20] are inequality conditions of monotonicity type for infinite systems. For finite systems, property (P) at the boundary point is the hypergroup property of the corresponding birth–death chain, and Miclo [10, Proposition 6 and Theorem 31], adapting arguments of Bakry and Huet, gives inequality conditions for Metropolis chains whose potential is sufficiently convex and either monotone or symmetric under $n \mapsto N - n$, and for chains symmetric under $n \mapsto N - n$ whose coefficients satisfy a monotonicity condition. These conditions do not cover the full Krawtchouk or Szwarc families. For symmetric Krawtchouk they fail when $N > 2$, but the cases $N = 1, 2$ are covered; the Szwarc instance $N = 1$, $p_0 = 1/2$ is the same covered three-point system. By the quadratic decomposition [12, Theorem 4.1], any criterion of this kind for nonsymmetric finite systems, valid including the terminal indices, immediately gives one for symmetric finite systems. Theorem 8.1 supplies such hypotheses for all triples except those whose pairwise sums all exceed N. Theorem 8.3 supplies such hypotheses for all triples under its strengthened conditions (i)–(iii), and Corollary 8.4 contains the arrays (4.2) with $q \geq 2$ but not the Krawtchouk arrays with $N \geq 3$. For constant-ratio arrays, the log-concave Newton criterion [12, Theorem 3.1] controls the deep triples by inequalities that contain the whole geometric criterion, in particular the Krawtchouk arrays, and [13] extends it to ratios $c_i/(b_i + c_i)$ that increase in several ways; other nonconstant ratios remain open. Find monotonicity-type conditions, valid through the terminal index, that control the deep triples and contain the Krawtchouk arrays with $N \geq 3$. Shape conditions alone do not suffice: convex profiles, convex profiles with log-concave or log-convex increments, and the sign classes of Appendix A contain arrays without property (P) at biases bounded away from 1, and Section 9 explains part of this at the symmetric endpoint.

Thinness along the terminal row. For every fixed $r \in (0, 1)$, [12, Corollary A.3] gives an exponential upper bound on the admissible coefficient-ball radius around $K_{N,r}$, witnessed by raising the top mass. Determine whether analogous upper bounds hold for terminal-row perturbations, and which uniform rules exhibit such thinness. The first-order heuristic of Appendix A changes at $r = \frac{1}{2}$, but the exact terminal-row failures recorded there for $r \leq \frac{1}{2}$ show that higher-order terms matter, so it does not locate a threshold for finite perturbations.

Linear inequality criteria. Decide whether a criterion given by linear inequalities in the coefficients can admit every $K_{N,r}$ and $G_{N,q,r}$ ($N \geq 1$, $q \geq 1$) for all r in a range $0 < r \leq r_0$; here $G_{N,q,r}$ is the array (4.2), so that $G_{N,1,r} = K_{N,r}$. Such a criterion admits a convex set of arrays, but the set of arrays with property (P) need not be convex, even at fixed (N, r) and with both endpoints admitted by Theorem 4.1: for $q > 1$ and $0 < \lambda < 1$ the combinations $(1 - \lambda)K_{N,r} + \lambda G_{N,q,r}$ have $g(N, N; N) < 0$ at seven exactly certified parameter choices with r between $2/7$ and $4/5$ [12, Section 5]. At those parameters no sufficient criterion whose admitted set is convex, in particular none defined by a system

of linear inequalities in the coefficients, can admit both $K_{N,r}$ and $G_{N,q,r}$. For smaller r the question is open; in our searches the examples found grew in size as r decreased (Appendix A). Sign conditions on the second differences of the profile and the first differences of the ratio fail for every $\rho_{\max} \geq 83/400$ (Appendix A). For $\rho_{\max} < 83/400$ the sign choices not excluded there remain unresolved, and criteria with conditions on higher differences, with mixed conditions, or with conditions that do not apply at every index are not covered.

Other finite families. Find coefficient regions around other finite orthogonal systems and compare them with the Krawtchouk neighborhoods. A positive algebra may supply the reference margin; a useful recurrence test must avoid computing the product coefficients.

A. Counterexamples and computations

All examples in this appendix are reproduced in the companion code [14]. Exact rational computations certify the finite examples of the first two paragraphs, and those listed in the supplement [16] are also machine-verified. The high-precision calculations, heuristics and searches of the last paragraph are numerical evidence only. No asymptotic conclusion is asserted. Here $g(N, N; N)$ is computed from column N of the product table, an $O(N^2)$-operation recurrence, so sizes in the hundreds are cheap.

Exact counterexamples to shape conditions. Each failure below is witnessed by a negative terminal coefficient $g(N, N; N)$, and all arrays have a constant ratio unless stated otherwise. The Lean declaration that decides each example in the kernel is given in parentheses (prefix `Szkraw.` omitted).

- *Convex profiles.* Convexity of the profile does not suffice, even with r bounded away from 1: for d with increments 1 at the first seven steps and 6/5 at the remaining seven ($N = 14$), $g(14, 14; 14) < 0$ at $r = 1/2$, and the analogous profile with $N = 20$, twelve unit increments and 21/20 afterwards, gives $g(20, 20; 20) < 0$ at $r = 2/5$ (`kink14_fails`, `kink20_2_5_fails`). One-kink profiles (m unit increments, then $N - m$ increments $1 + \delta$) give $g(N, N; N) < 0$ at

r	2/5	1/3	3/10	1/4	2/7	1/5	1/10
N	20	40	40	60	80	100	500
m	12	28	28	45	60	80	450
δ	1/20	1/5	1/10	1/10	1/5	1/10	1/10

  (`kink20_2_5_fails`, `kink40_1_3_fails`, `kink40_3_10_fails`, `kink60_1_4_fails`, `kink80_2_7_fails`, `kink100_1_5_fails`, `kink500_1_10_fails`, in the order of the table). Below $r = \frac{1}{2}$ the counterexamples we found occur at sizes that grow as r decreases; we did not determine minimal sizes or prove a scaling.

- *Log-convex increments.* Fourteen unit increments followed by $(21/20)^t$, $t = 1, \ldots, 4$, give $g(18, 18; 18) < 0$ at $r = 1/2$ (`logconvex18_fails`).

- *Log-concave increments.* Convex profiles with log-concave increments (step ratios nonincreasing and at least 1) contain Krawtchouk, every geometric profile and the normalized quadratic profiles $d_\kappa(x) = (x + \kappa x^2)/(1 + \kappa)$, $\kappa \geq 0$, and they exclude the one-kink counterexamples. Yet k steps of ratio $1 + 1/(2N)$ followed by flat increments give $g(N, N; N) < 0$ at $(r, N, k) = (\frac{1}{2}, 80, 40)$, $(\frac{2}{5}, 32, 22)$ and $(\frac{1}{3}, 90, 62)$ (`logconcave80_fails`, `logconcave32_fails`, `logconcave90_fails`).

- *Sign classes.* Write $d_n = b_n + c_n$ and $\rho_n = c_n/d_n$ for $n \geq 1$. For $S_1, S_2 \in \{[0, \infty), (-\infty, 0], \{0\}, \mathbb{R}\}$ and $\rho_{\max} > 0$, consider the stochastic arrays with $b_0 = 0$, a_n decreasing, b_n and c_n nondecreasing, $d_{n+2} - 2d_{n+1} + d_n \in S_1$ for $0 \leq n \leq N - 2$, $\rho_{n+1} - \rho_n \in S_2$ for $1 \leq n \leq N - 1$, and $\rho_n \leq \rho_{\max}$. Each such class contains every $K_{N,r}$ with $r \leq \rho_{\max}$, and for $S_1 = S_2 = \{0\}$ it consists of these arrays only. For every other choice of (S_1, S_2) and every $\rho_{\max} \geq 83/400$, exact computation [14] gives an array in the class with $g(N, N; N) < 0$. In the first two examples, normalize the listed increments to sum to one. If S_1 contains a positive number, take constant ratio $1/5$ and eighty unit increments followed by twenty increments $11/10$ ($N = 100$). If S_1 contains a negative number, take constant ratio $1/10$ and seventeen unit increments followed by three increments $1/2$ ($N = 20$). If $S_1 = \{0\}$ and S_2 contains a positive number, take $d_n = n/N$ and $\rho_n = 1/5 + \max(n - 102, 0)/2400$ ($N = 120$). If $S_1 = \{0\}$ and S_2 contains a negative number, take $d_n = n/N$ and $\rho_n = 1/5 - \max(n - 68, 0)/1600$ ($N = 80$). Thus, for every $\rho_{\max} \geq 83/400$, among the classes just defined, only $S_1 = S_2 = \{0\}$ guarantees property (P) for all N (`sign_pattern`). For $\rho_{\max} < 83/400$, these four examples do not settle all sign choices; the cases not excluded by them remain unresolved here.

These examples disprove the stated shape conditions at the listed parameters and sizes.

Exact failures along deformation directions.

- For the rule $\beta = 1$, $\gamma = 0$, exact computation shows that property (P) fails at $r = 4/5$, with perturbation size $1/100$, for every $N = 6, \ldots, 14$, and at $r = 19/20$, with perturbation size $1/1000$, for every $N = 4, \ldots, 14$. The period-four sign pattern at $r = 1$ does not persist in general for $r < 1$: at $r = 4/5$ the first-order derivatives are $512/1875$ for $N = 4$ and $-71936/5859375$ for $N = 8$ (`rule_fails_4_5`, `rule_fails_19_20`, `PolyCol.rule_deriv_4_5`).

- Along the terminal row, where $g(N, N; N)$ is a polynomial in the perturbation, exact computations confirm failures of property (P) at sizes down to 10^{-15} for $r = 4/5$, $N = 40$. Failures also occur for $r \leq \frac{1}{2}$: at size $6.3 \cdot 10^{-6}$ for $r = \frac{1}{2}$, $N = 120$; at size $3.2 \cdot 10^{-3}$ for $r = \frac{2}{5}$, $N = 100$; and at size $1.6 \cdot 10^{-2}$ for $r = \frac{1}{3}$, $N = 120$ (`terminal_fails`).

Numerical evidence and heuristics. None of the following is an exact sign certificate.

- In 80- and 150-digit floating-point arithmetic, which agree, the log-concave-increment construction above also gives $g(N,N;N) < 0$ at $(r,N,k) = (\frac{1}{4},250,175)$ and $(\frac{1}{5},500,400)$.

- In 150- and 300-digit arithmetic, the combination $(1-\lambda)K_{N,r} + \lambda G_{N,q,r}$, whose increments are proportional to $1 + t\,q^{i-N}$ with $t = \lambda N(q-1)q^{N-1}/((1-\lambda)(q^N-1))$, has $g(N,N;N) < 0$ at $(r,N,q,t) = (1/4,500,3,10)$. At $r = 1/5$ no negative corner was found on the tested (N,q,t) grid with $N \le 1000$.

- For the sign class with $S_1 = \{0\}$ and increasing ratio starting from $\rho = 1/10$, our search found no failure for $N \le 300$.

- Along the terminal row alone, the first-order change of $g(N,N;N)$ at $K_{N,r}$ has an exact contour-integral representation, obtained as in Lemma 9.2 with $R_N(J)$ having eigenvalues $(-r)^j$. The formal saddle contributions have exponential factor $(1-r+r^2)/(1+r)$, which exceeds $1-r$ exactly when $r > \frac{1}{2}$. For $r > \frac{1}{2}$ this suggests an oscillatory first-order envelope, not a proved threshold for finite perturbations; no asymptotic estimate or uniform control of the nonlinear remainder is proved here.

- Floating-point searches for Szwarc-type certificates on the deep triples are recorded in the companion code.

These findings establish neither minimal failure sizes nor general conclusions about smooth deformations.

B. Known positivity and Koelink's construction

This appendix derives the classical positivity results from Theorem 4.1 and compares the geometric family with Koelink's construction. The positivity statements are known; they are included because Section 3 uses the Krawtchouk case as its reference.

Corollary B.1 *(Full positive Krawtchouk range).* *For every $N \ge 1$ and $0 < p_0 \le 1/2$, Krawtchouk polynomials orthogonal for binomial weights $\binom{N}{j}p_0^{N-j}(1-p_0)^j$ have nonnegative linearization when normalized at $j = 0$.*

Proof. To fix the convention without relying on a named recurrence, put $r = p_0/(1-p_0) \in (0,1]$, and let $Y_1,\dots,Y_N$ be independent variables with values $1, -r$ of probabilities $p_0, 1-p_0$. Their mean is zero and $Y_i^2 = (1-r)Y_i + r$. The normalized elementary symmetric sums

$$K_i(j) = \binom{N}{i}^{-1} \sum_{|S|=i} \prod_{l\in S} Y_l$$

depend only on the number j of negative coordinates. They are orthogonal, because the expectation of two distinct squarefree monomials is zero: some independent mean-zero variable occurs once. Their weights in j are exactly the stated binomial weights. With

$x = N^{-1}\sum_l Y_l = 1 - j/(N(1-p_0))$, multiplication by x gives

$$xK_i = \frac{N-i}{N}K_{i+1} + (1-r)\frac{i}{N}K_i + r\frac{i}{N}K_{i-1}.$$

Indeed, in multiplying the elementary sum by $\sum_l Y_l$, indices outside a subset give $(i+1)$ copies of each subset of size $i+1$; the linear part of Y_l^2 gives $(1-r)i$ copies of each subset of size i; and the constant part gives $r(N-i+1)$ copies of each subset of size $i-1$. Division by the binomial normalizers gives the displayed recurrence, including $i = N$. Its leading coefficient in x is positive, its degree is i, and its value at $j = 0$ is one. These properties identify the normalized orthogonal polynomials for the binomial measure. Apply Theorem 4.1 with $q = 1$. The range is sharp in this endpoint normalization: if $1/2 < p_0 < 1$, then $r = p_0/(1-p_0) > 1$ and $g(1,1;1) = b_1 = (1-r)/N < 0$. For $N = 1$, this coefficient follows from the terminal relation. ■

Corollary B.2 *(The dual q-Krawtchouk case).* *For every $N \geq 2$, $q > 1$, and $0 < r \leq 1$, the recurrence (4.2) has nonnegative linearization. In the normalization $R_0 = 1$, $R_1 = x$, it is outside the Krawtchouk recurrence class.*

Proof. Its increments are $d_i - d_{i-1} = q^{i-1}/[N]_q$, so all hypotheses hold. For a Krawtchouk recurrence in this normalization the increments of $b_i + c_i$ are constant. Here their ratio is $q > 1$ and $N \geq 2$, so they are not constant. An affine change of the support transporting the same endpoint-normalized basis leaves the multiplication coefficients in the basis unchanged; it cannot remove this distinction. ■

Proposition B.3 *(Relation to the existing algebraic construction).* *The quadratic relations (5.1) are those of Koelink [8, Theorem 3.1(ii), relation (3.2)], with $z_j = u(x^j)$ and the same parameters p, q. The recurrence in this paper is the normalized recurrence of his Theorem 7.1, with $r = 1/p$; its polynomial-in-the-spectral-variable interpretation is the dual q-Krawtchouk interpretation described after his Theorem 7.8.*

Proof. The source square relation is $u(x^j)^2 = (q-1)u(x^j)\sum_{l>j} u(x^l) + (p-1)u(x^j) + pq^{N-j}$, which is (5.1). In Theorem 7.1 the coefficients of the next, current and previous indices are respectively $pq^i(1-q^{N-i})$, $(p-1)(1-q^i)$ and $1-q^i$. Division by $p(1-q^N)$ gives (4.2) for $q > 1$. For $0 \leq f \leq N$, the corresponding spectral coordinate is

$$x_f = \frac{p(1-q^{N-f}) - (1-q^f)}{p(1-q^N)} = \frac{q^{N-f} - rq^f + r - 1}{q^N - 1}, \qquad r = 1/p.$$

Writing $\phi_f(w_d)$ for Koelink's spherical-function values, we have $R_d(x_f) = \phi_f(w_d)$ for $0 \leq d \leq N$: the initial value is one and the normalized recurrences agree, so induction in d identifies them. As $q \to 1$, dividing numerator and denominator by $q-1$ and using the finite geometric sums gives $x_f \to 1 - (1+r)f/N$. The $q = 1$ case follows directly from the construction above, without division by zero. His discussion after Theorem 7.8 explicitly interchanges polynomial degree and spectral index to obtain the dual family. ■

The geometric criterion covers the full positive Krawtchouk range, including the symmetric endpoint. The perturbation criterion allows freely varying coefficients around each Krawtchouk array in the interior of the positive linearization range; its stated radius is conservative and decreases with N. Both hypotheses can be checked using a number of arithmetic comparisons proportional to N, without forming any general product coefficients or testing polynomial matrices. The proofs use positive algebra multiplication, a lower bound from a single positive term, and a norm estimate. They provide sufficient conditions, not a characterization of all finite systems with property (P). The geometric parameter family and underlying algebra predate Szwarc's survey; this paper makes no claim of historical priority for these constructions or criteria.

AI-use disclosure

GPT-6 Astra or GPT-6.1 Sol (OpenAI; which of the two was not recorded), run in the Codex agent; GPT-6.1 Sol, run as a separate Codex reviewing agent; Claude Opus 5.5 (Anthropic), run in the Claude Code agent; and GPT-6 Pro (OpenAI), used in ChatGPT, were used to develop the proofs, the exact checks, the Lean formalization and the text. The GPT-6.1 Sol agent and GPT-6 Pro reviewed the results. This AI review is not human peer review, and no independent human verification is claimed. The author set the research objective and is responsible for the mathematical claims. The supplement [16] gives the full disclosure, including which sections arose with which model.

Code and formalization

All theorems, lemmas, propositions and corollaries are formalized in Lean 4 with Mathlib in the companion code [14], whose axiom audit admits only `propext`, `Classical.choice` and `Quot.sound`; "machine-verified" refers only to these Lean theorems, and the exact finite counterexamples of Appendix A are decided in the Lean kernel. Two Lean proofs take a route different from the written one and use the Lean development of [12]: Theorem 7.1 is proved through the parity-split criterion [12, Corollary 4.2] with $q = 1$, and Lemma 7.4 through the quadratic decomposition [12, Theorem 4.1]. The supplement [16] lists the Lean declaration for each result and remark, describes the treatment of the largest finite certificate, and lists what is not formalized.

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